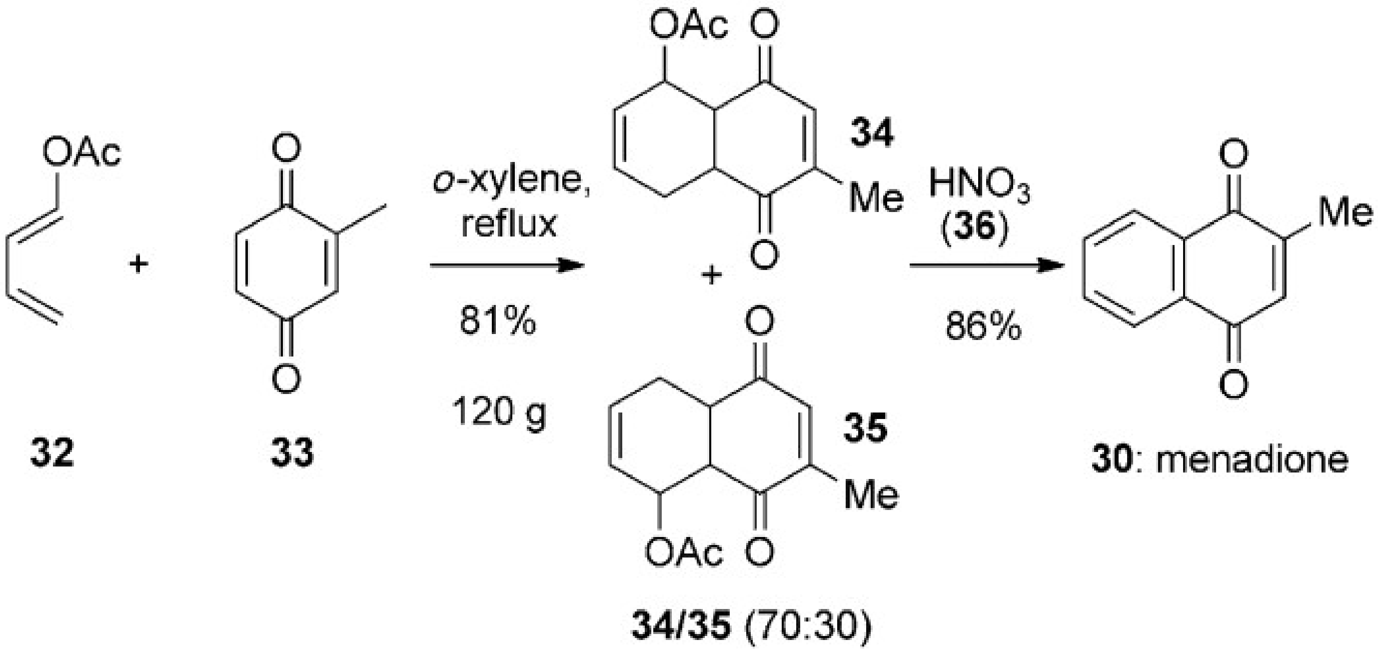



# Preparation of provitamin $K_3$

# PERICYCLIC RACTION

- Pericyclic reaction involves a cyclic redistribution of bonding electron through a concerted process.
- Concerted process means bond breaking and making occurs in a single step.

cycloaddition

Electrocyclic

Pericyclic reaction

Sigmatropic

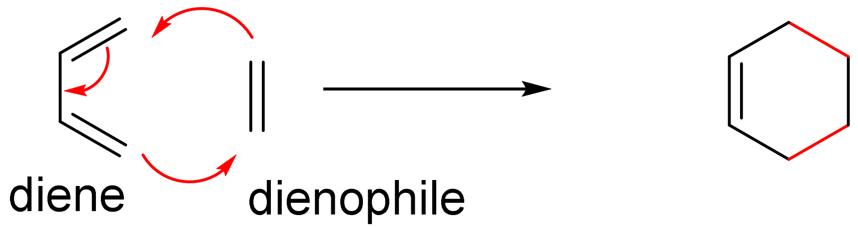
Group Transfer

# 1. Cycloaddition Reaction

- Formation of a new ring.
- More than one bond form simultaneously.
- No intermediate formation.
- Three important classification of cycloaddition reaction
  - I. [4+2]-cycloaddition/Diels-Alder reaction
  - II. [1,3]-Dipolar cycloaddition
  - III. [2+2] cycloaddition

# Diels-Alder reaction

- Reaction between a conjugated diene and dienophile.
- Two π bonds disappear and two new σ bond formed.
- This is a [4+2] cycloaddition.
- Diels Alder reaction takes place under thermal condition.

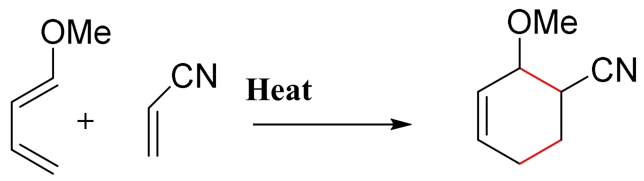


## Classification of Diels-Alder reaction:

- Normal [4+2]

  Diene is electron-rich

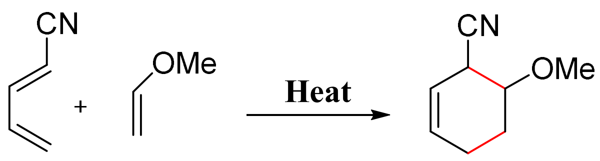
  Dienophile is electron-poor




- Inverse electron-demand [4+2]

  Diene is electron-poor

  Dienophile is electron-rich

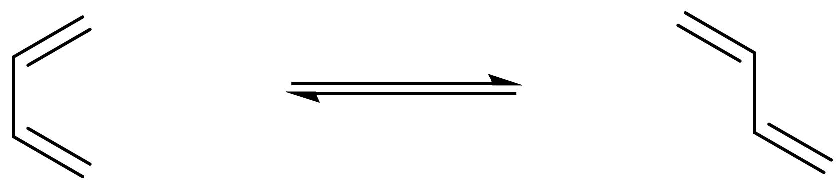



- Hetero[4+2]

  Hetero atom can be a part of diene or Dienophile or both

## Requirements of Diene:

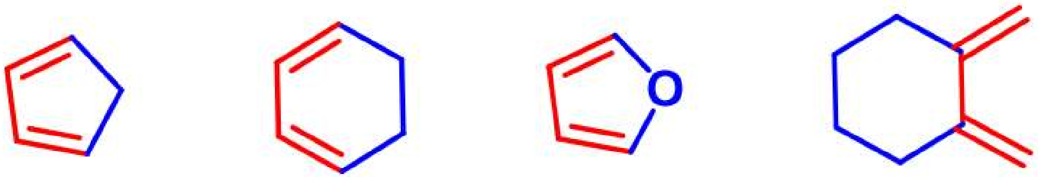
- Diene can be open chain or cyclic.
- Diene should be electron rich  and reactivity can be enhanced by electron donating group.
- Acyclic Diene must be s-cis conformation to be reactive.



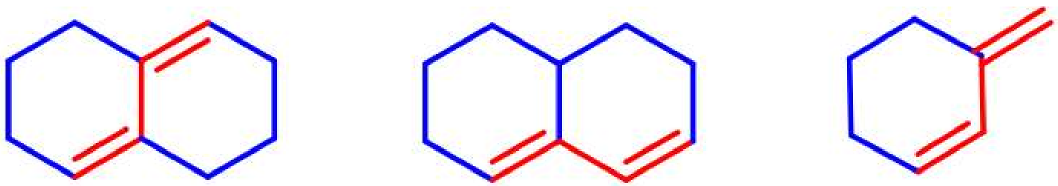
s-cis
(reactive intermediate)

s-trans
(Unreactive conformation)

- Cyclic diene which adopt s-cis conformation to be very reactive.

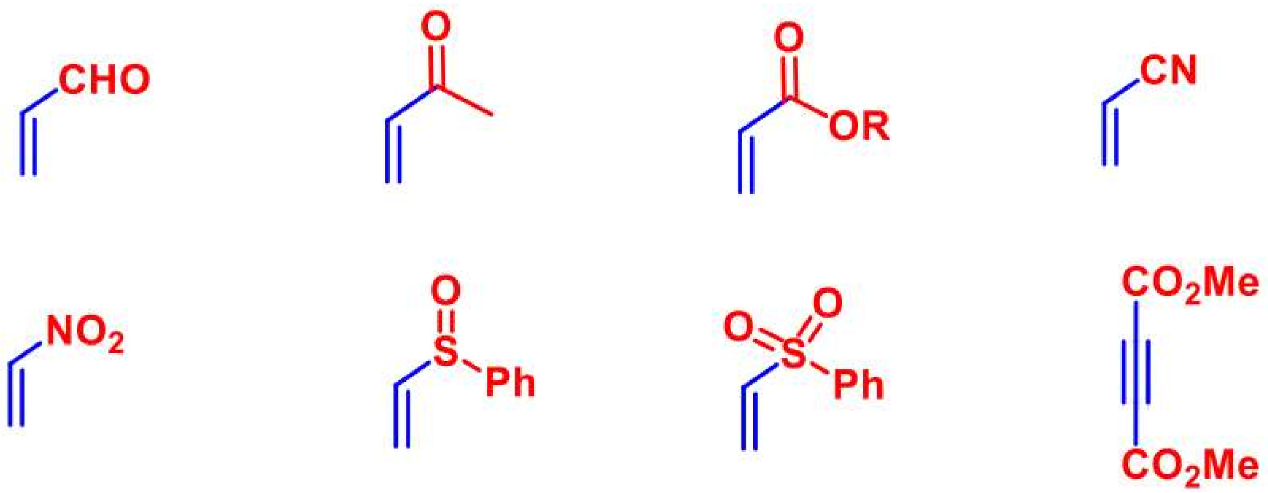


- Cyclic diene which are s-trans conformation are unreactive in Diels-Alder reaction.



## Requirements of Dienophile:

- Dienophile can be open or cyclic.
- Dienophile should be electron deficient and reactivity enhanced by electron withdrawing substituents

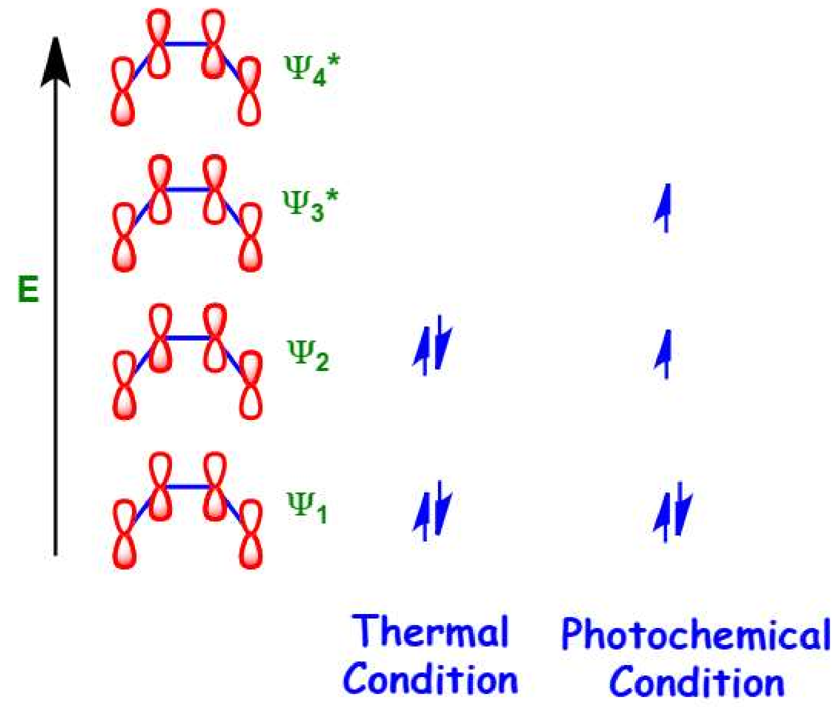


# Frontier Molecular Orbital (FMO) Approach

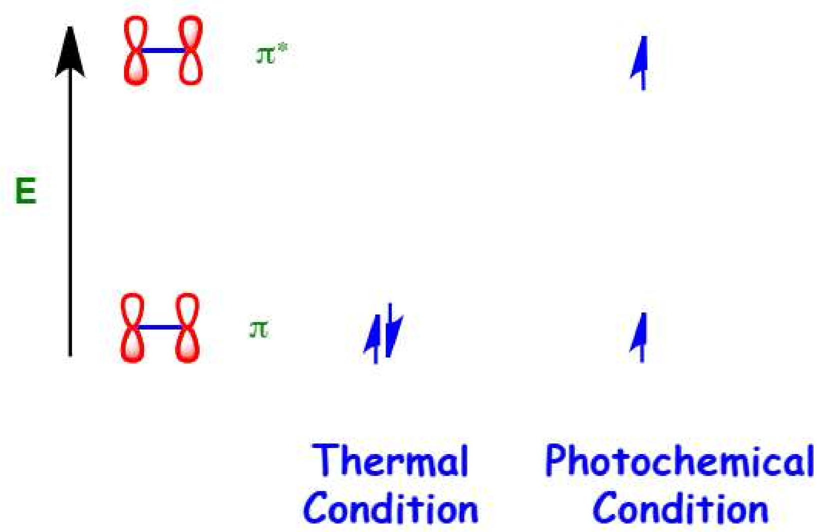
- Highest occupied Molecular Orbital (HOMO) & Lowest Unoccupied Orbital (LUMO) are known as Frontier Molecular Orbitals.
- Occupied orbitals repel each other.
- HOMO and LUMO interact with each other causing attraction.
- FMO simplifies the reactivity by interaction between HOMO of one & LUMO of other.

## Frontier Molecular Orbital (FMO) approach:

MO of Diene

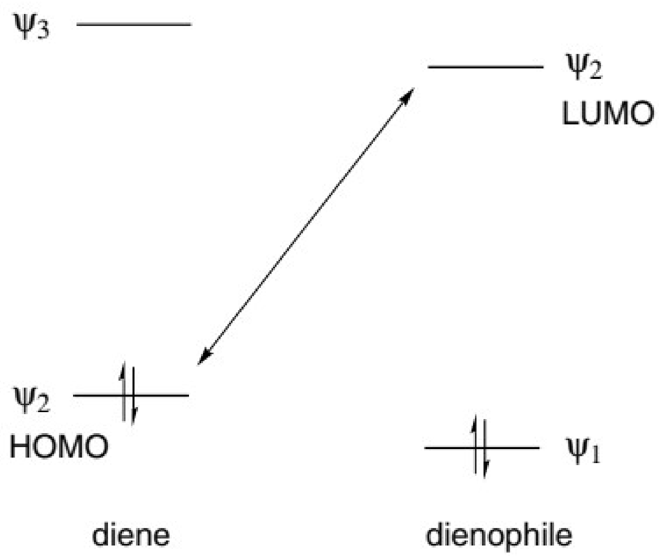


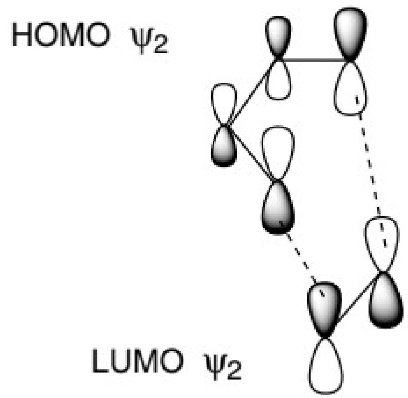

MO of Dienophile

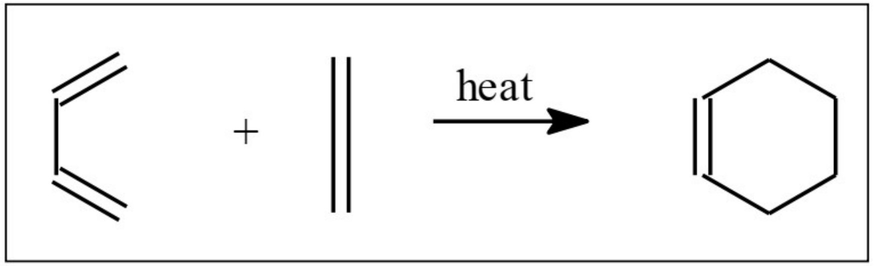


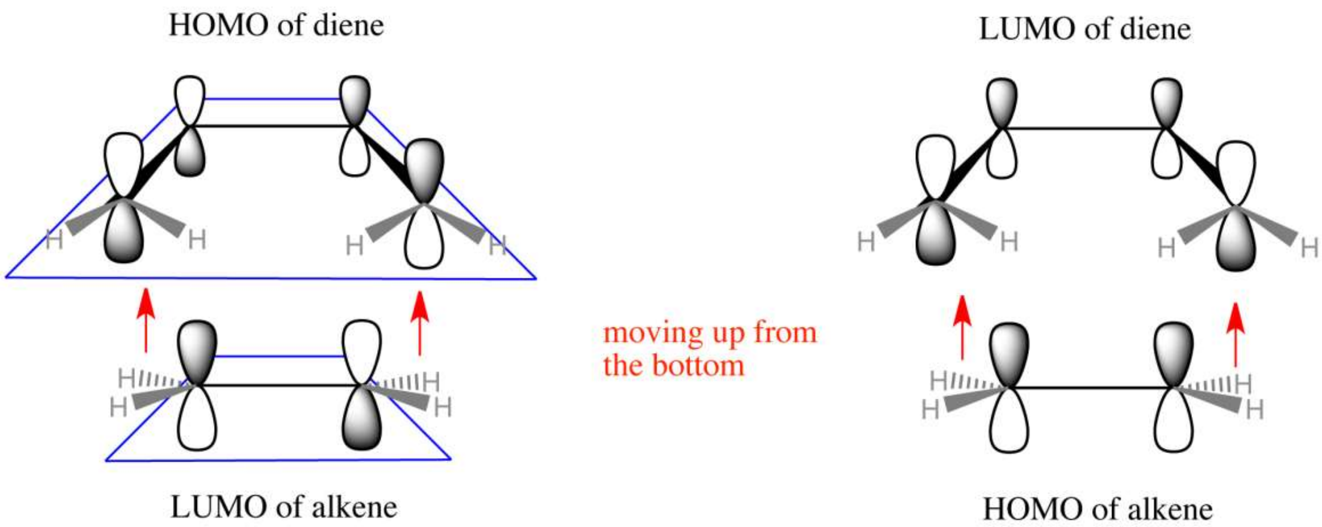
- A normal Diels Alder reaction  involve involves an electron diene and electron deficient dienophile.

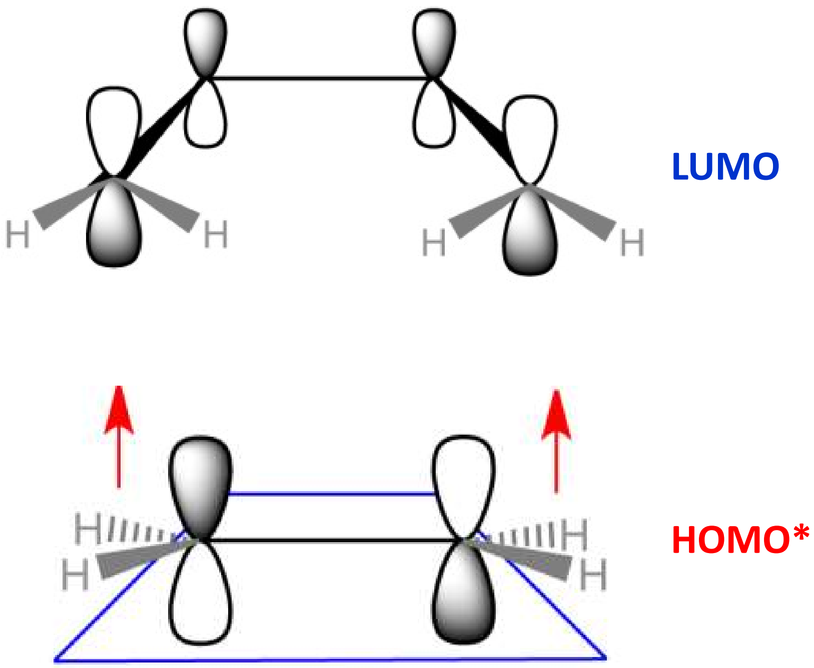
- The smallest the energy gap between these frontier orbitals, the better interact and more readily the reaction occur.

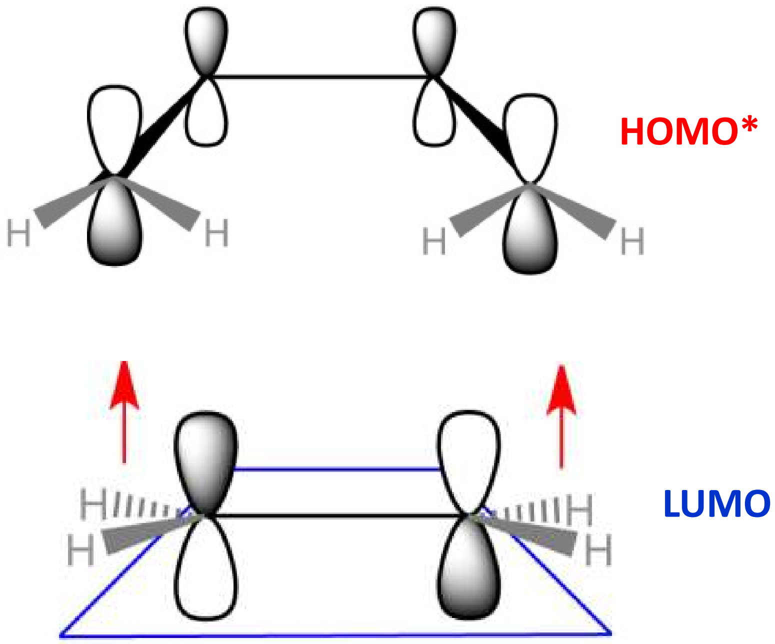



heat

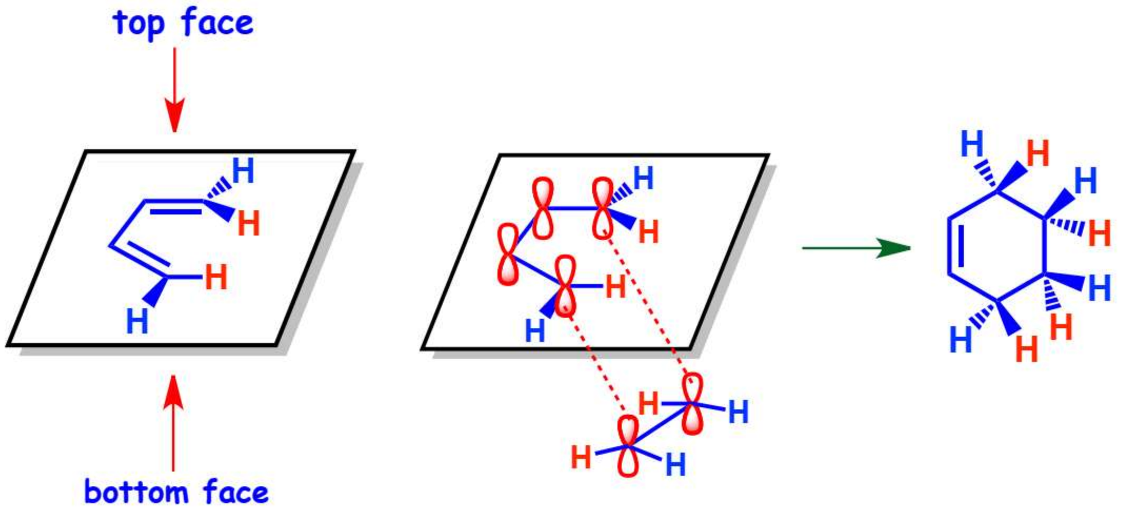

HOMO of diene
LUMO of diene
moving up from
the bottom
LUMO of alkene
HOMO of alkene


LUMO
HOMO*


HOMO*
LUMO

# Stereochemistry of Diels – Alder Reaction

- Diels-Alder reaction is stereospecific.
- Stereochemistry of both diene and dienophile retained in the product.
- Reaction controls the relative stereochemistry at four contiguous centres.
- Diels-Alder reaction occurs due to the overlapping of p-orbitals of diene and dienophile lying perpendicular to the plane of carbon atoms.
- Dienophile can approach either of the faces and lead to racemic mixtures.



# Stereochemistry of Diene

## Rule of Thumb-

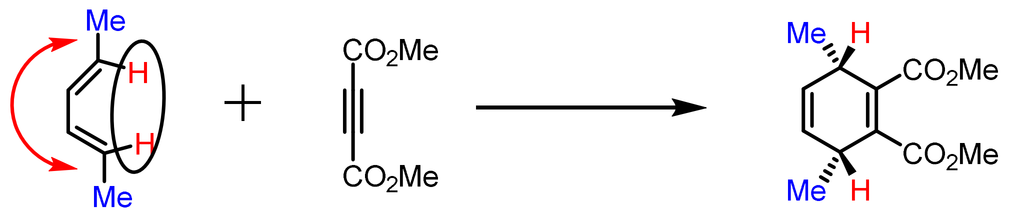
- Given the bottom face approach of dienophile:

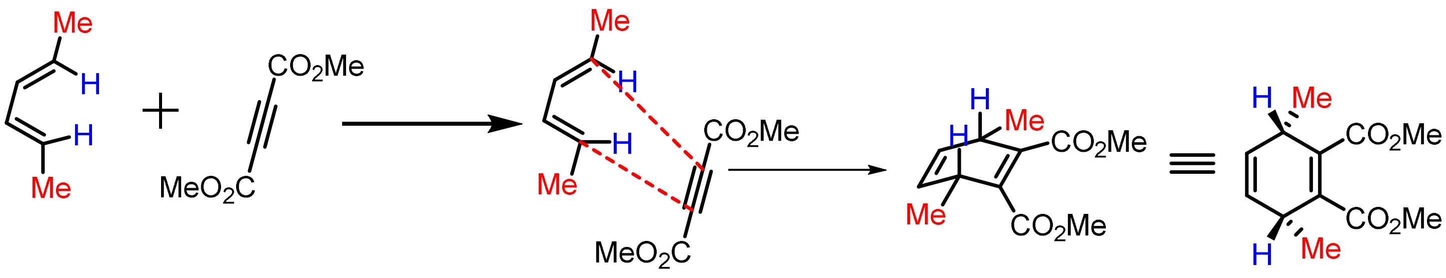
  -Inside substituents of diene will become β-configuration in the product.

  -Outside substituent of diene will become α-configuration in the product.

Inside substituents

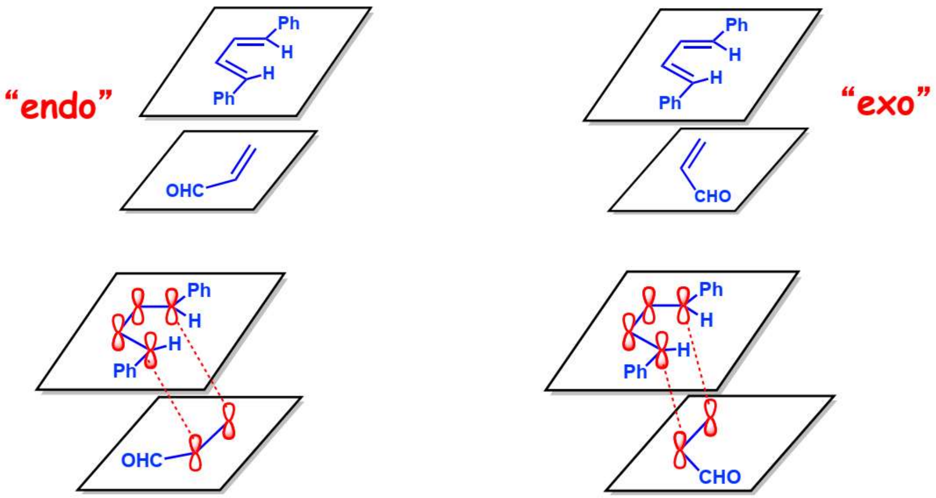
Outside substituents



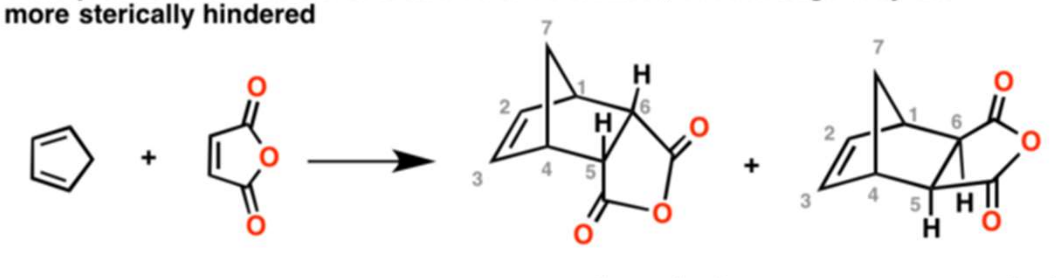



# Stereochemistry of Dienophile

- Given the bottom face approach of dienophile, substituents on dienophile can take different orientations with respect to the plane of the diene called endo or exo.




*endo* products tend to be favored in the Diels-Alder, even though they are more sterically hindered

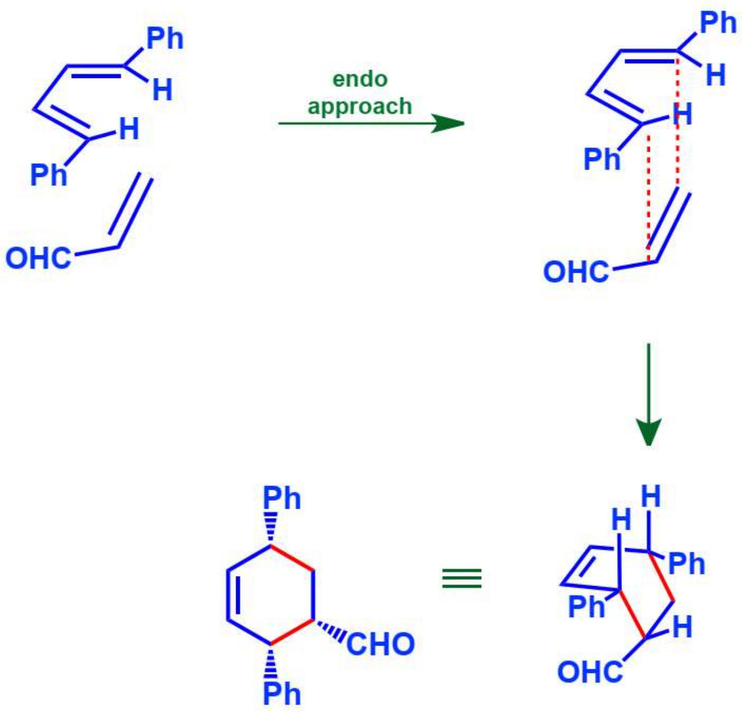


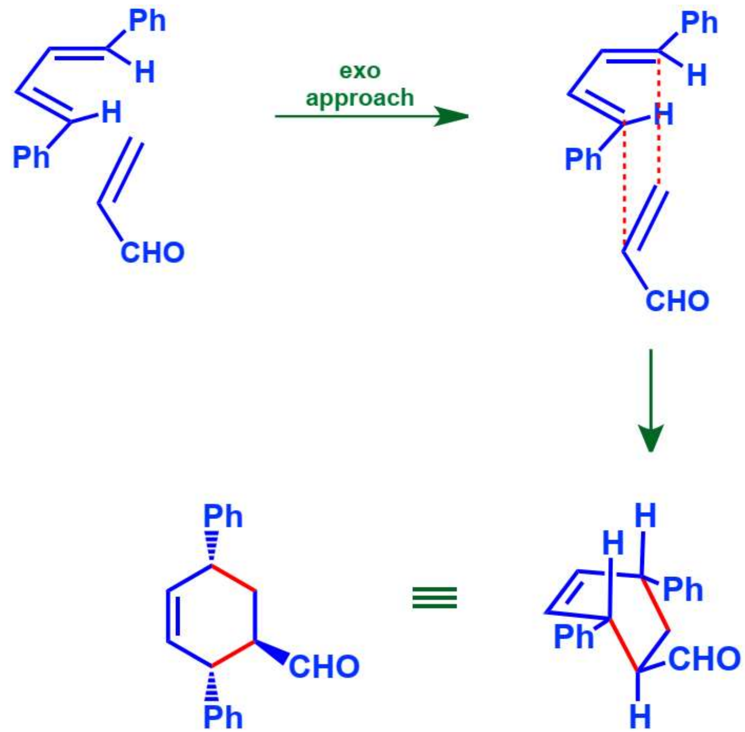

*endo* product

*exo* product

*endo: exo* = 4: 1

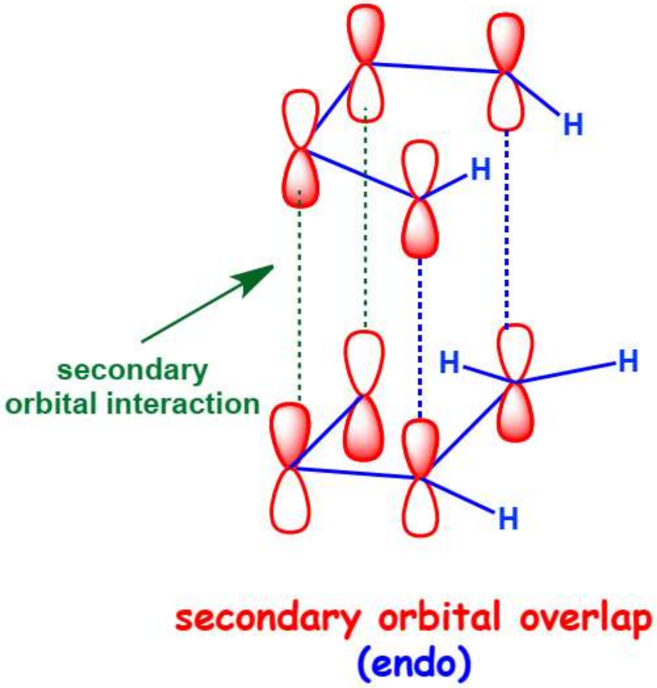
Example-

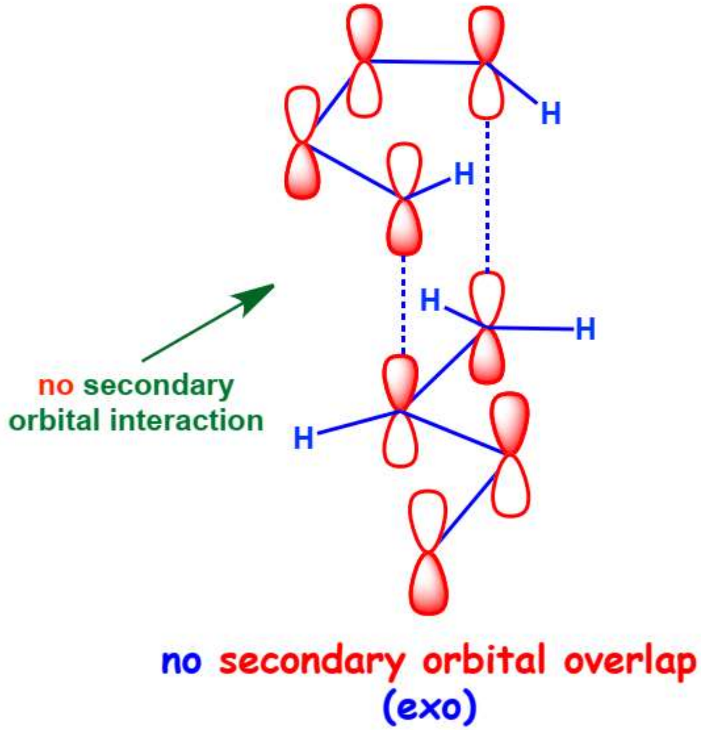


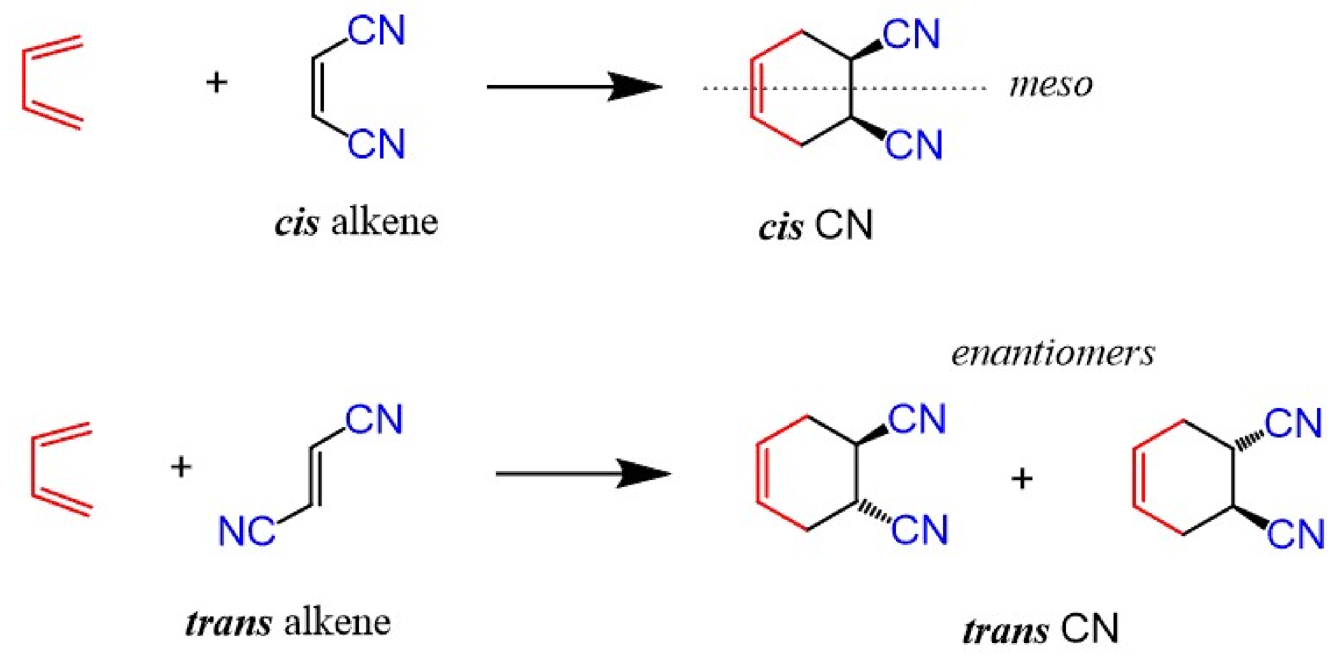
# Alder's endo Rule

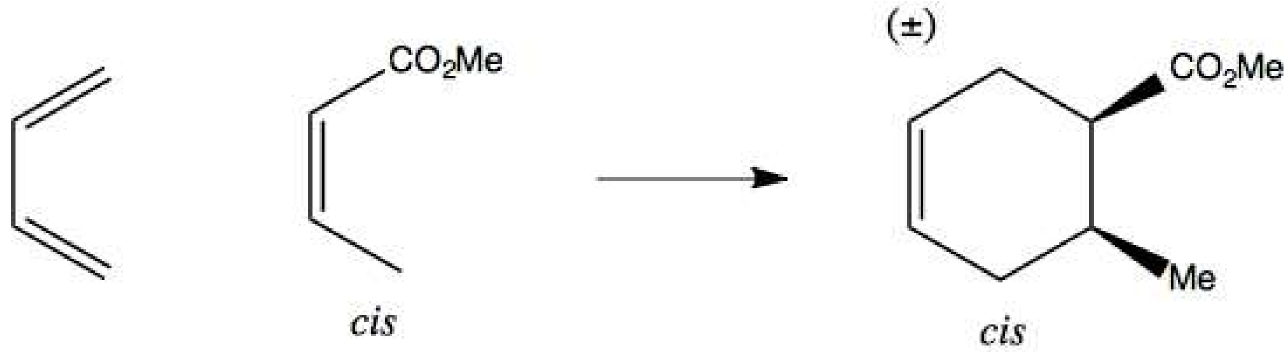
➢ Endo mode of activation is usually preferred to exo addition due to secondary orbital overlap between the dienophile's activating substituent and the diene.

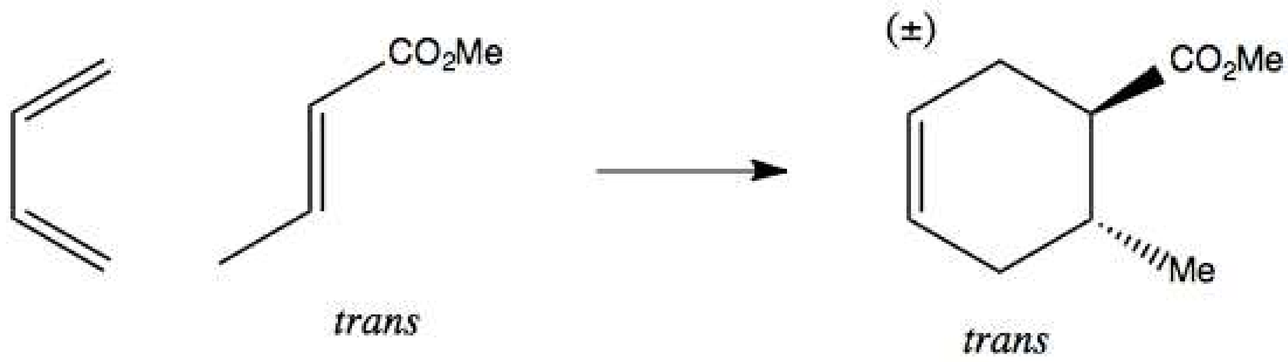


**Stereospecific reaction - *cis* produces *cis, trans* produces *trans***

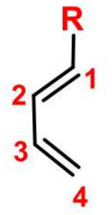



CO2Me
cis
(±)
CO2Me
Me
cis

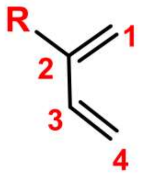

CO2Me
trans
(±)
CO2Me
Me
trans

# Regiochemistry of Diels – Alder Reaction

- For a mono- substituted diene, outcome of the product depends on carbon atom where it is substituted.
- Consider two classical cases, where the electron donating substituent could be present at carbon 1 and 2.

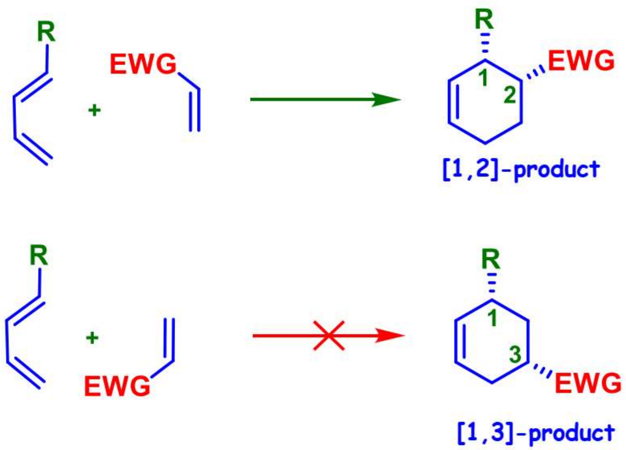



1-substituted diene

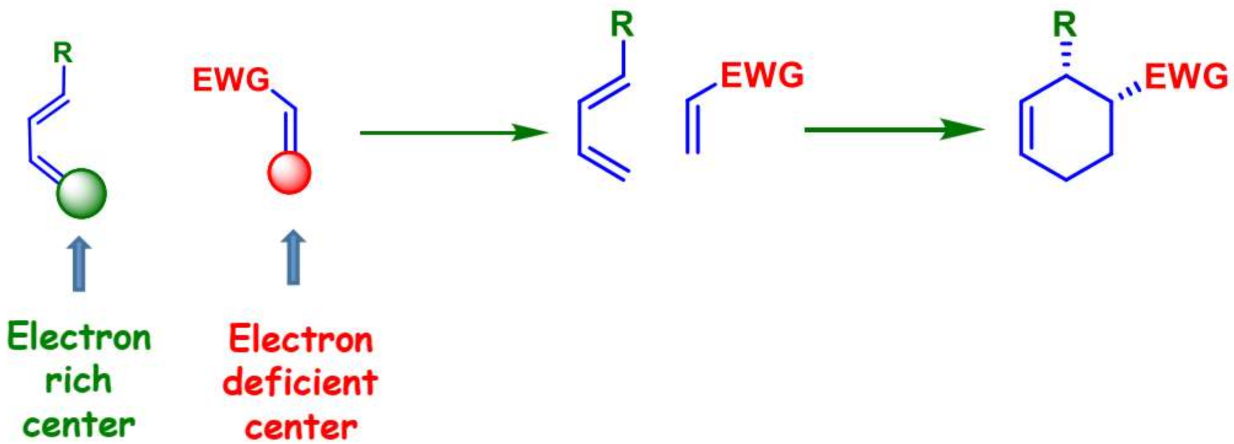



2-substituted diene

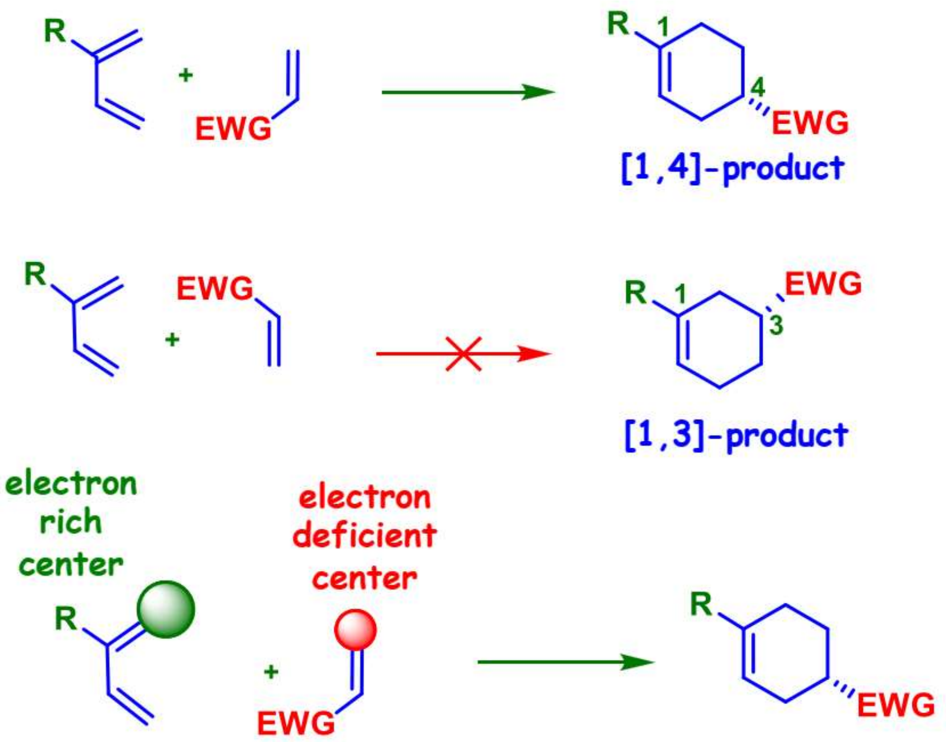
- 1- substituted diene react to give mainly [1,2]-product.



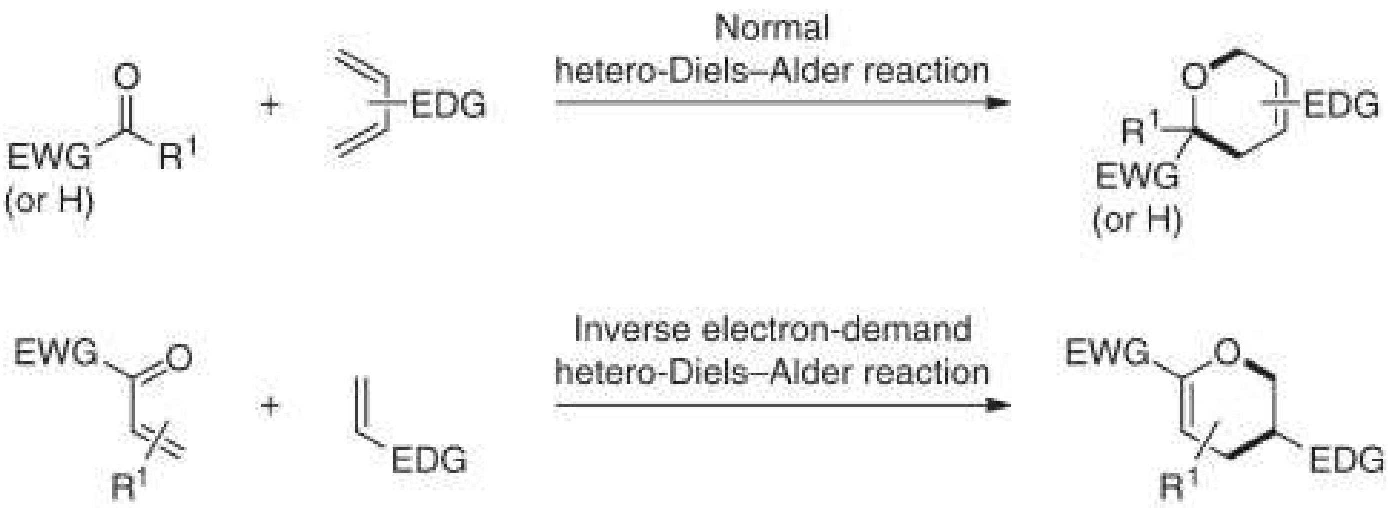
- ➢ Regioselectivity for the preference of [1,2]- product can be explained by considering the polarization of diene and dienophile.
- ➢ Connect the electron rich centre of diene to electron deficient centre of dienophile.



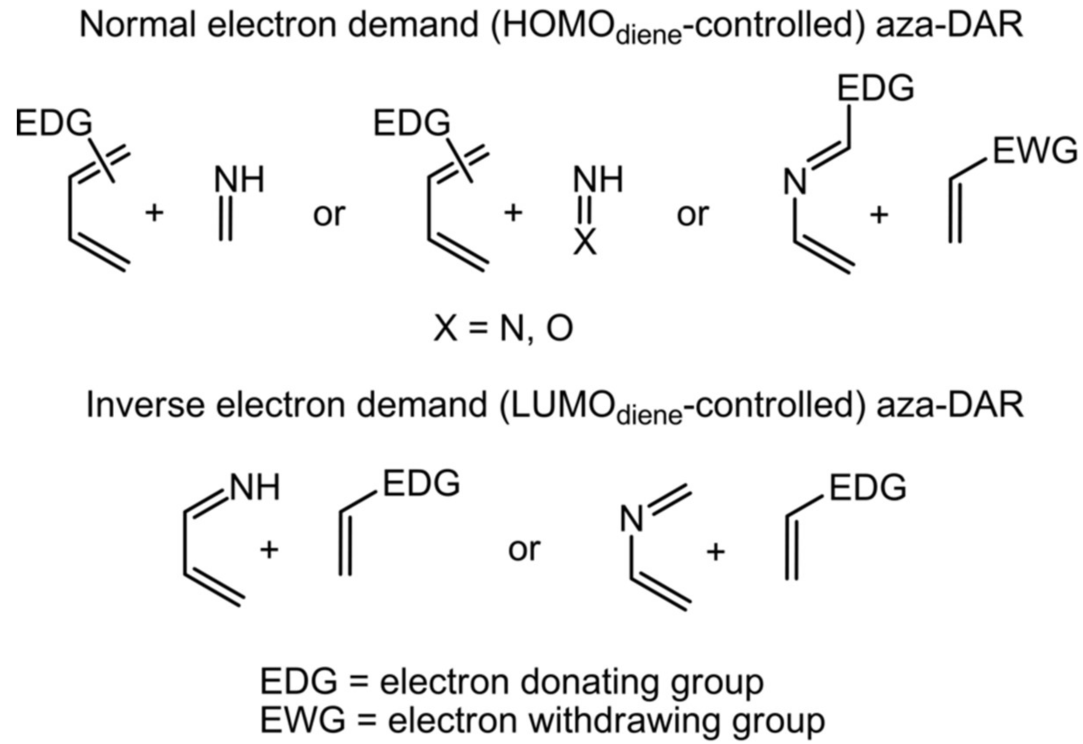
➢ 2- substituted diene react to give mainly [1,4]- substituted product.



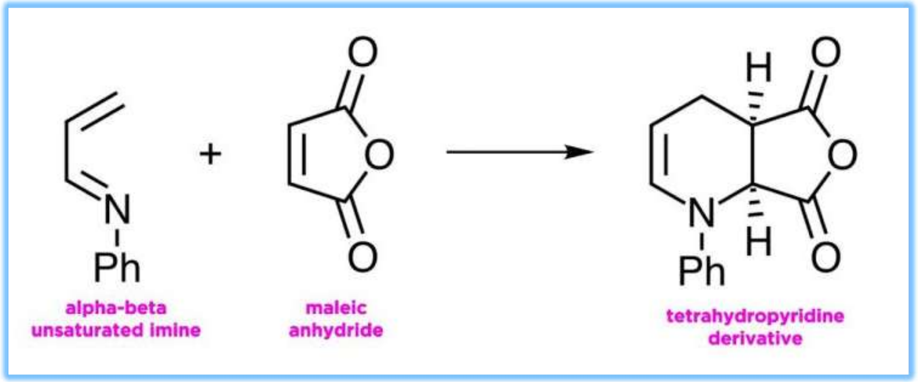
# hetero-Diels-Alder reaction



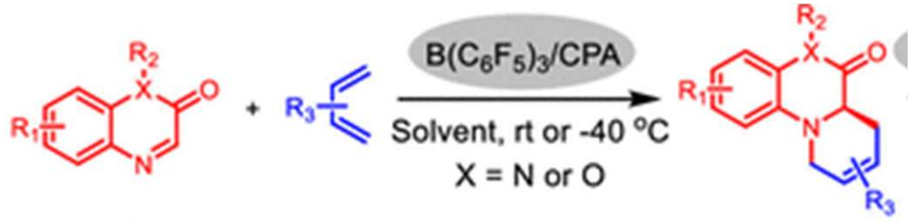

EDG, electron-donating group; EWG, electron-withdrawing group



Normal electron demand (HOMOdiene-controlled) aza-DAR
EDG
NH
or
X
X = N, O
EWG
Inverse electron demand (LUMOdiene-controlled) aza-DAR
EDG = electron donating group
EWG = electron withdrawing group

alpha-beta unsaturated imine
maleic anhydride
tetrahydropyridine derivative

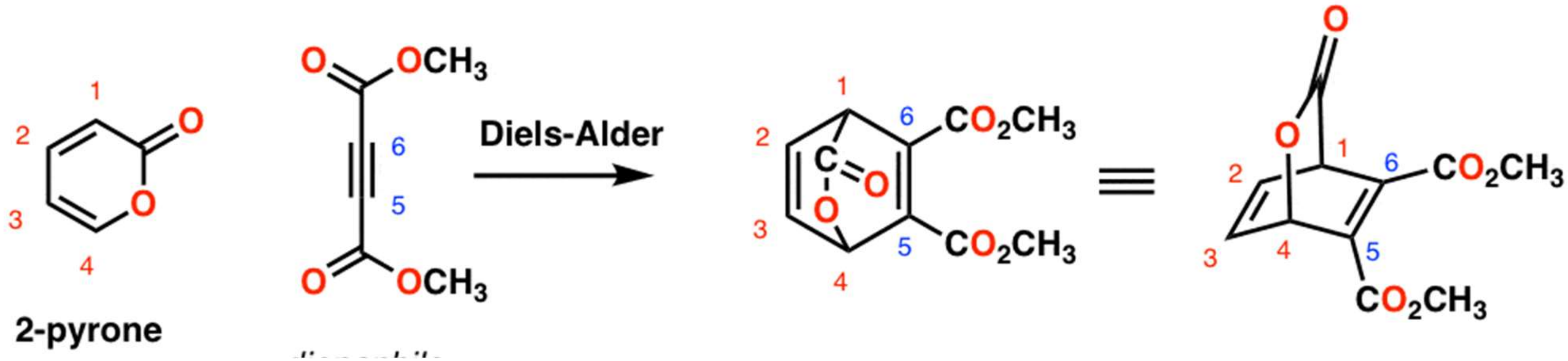



B(C6F5)3/CPA
Solvent, rt or -40 °C
X = N or O

# Retro Diels-Alder Reaction

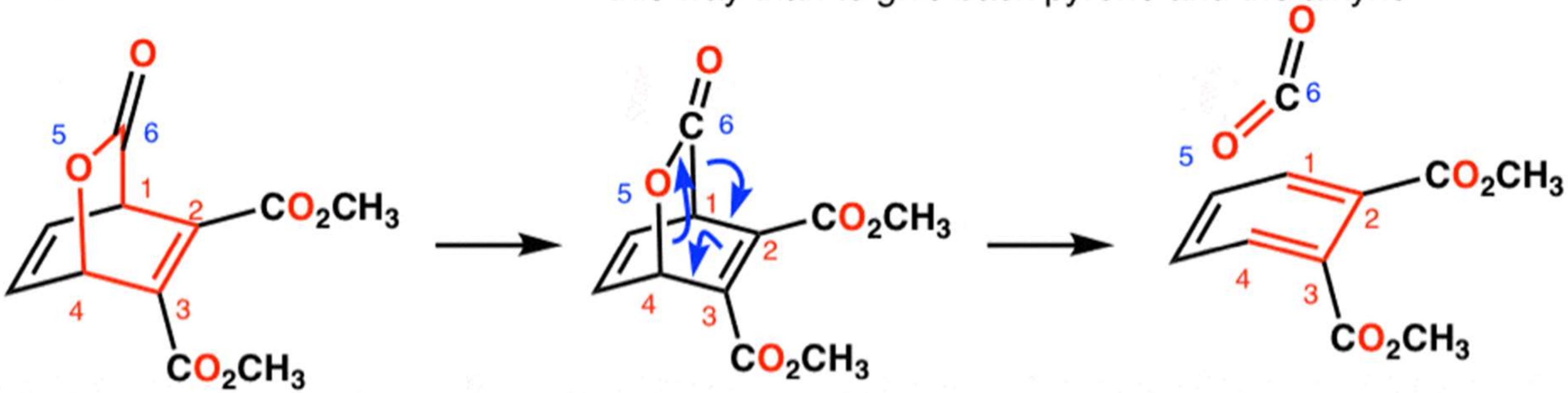
Diels-Alder product can undergo **more than one** possible *retro*-Diels Alder reaction

**2-Pyrone as a diene in the Diels-Alder with an electron-poor alkyne dienophile:**



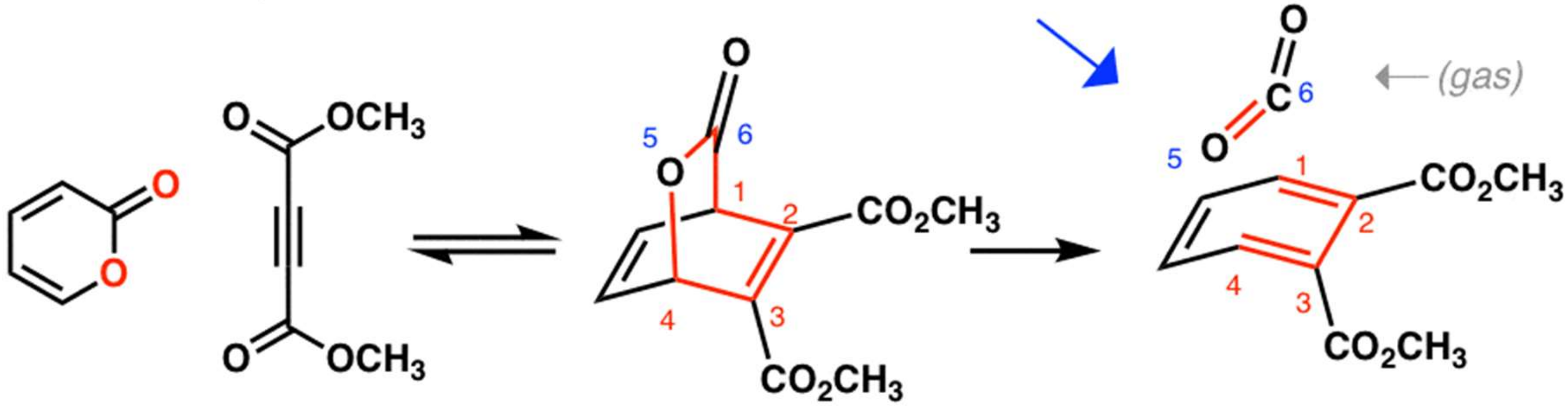
**The 6 membered ring highlighted in red undergoes retro Diels-Alder fragmentation**

- The products of this retro-Diels alder are $CO_2$ gas and a (very stable) aromatic ring
- More favorable (and irreversible) to fragment this way than to give back pyrone and the alkyne



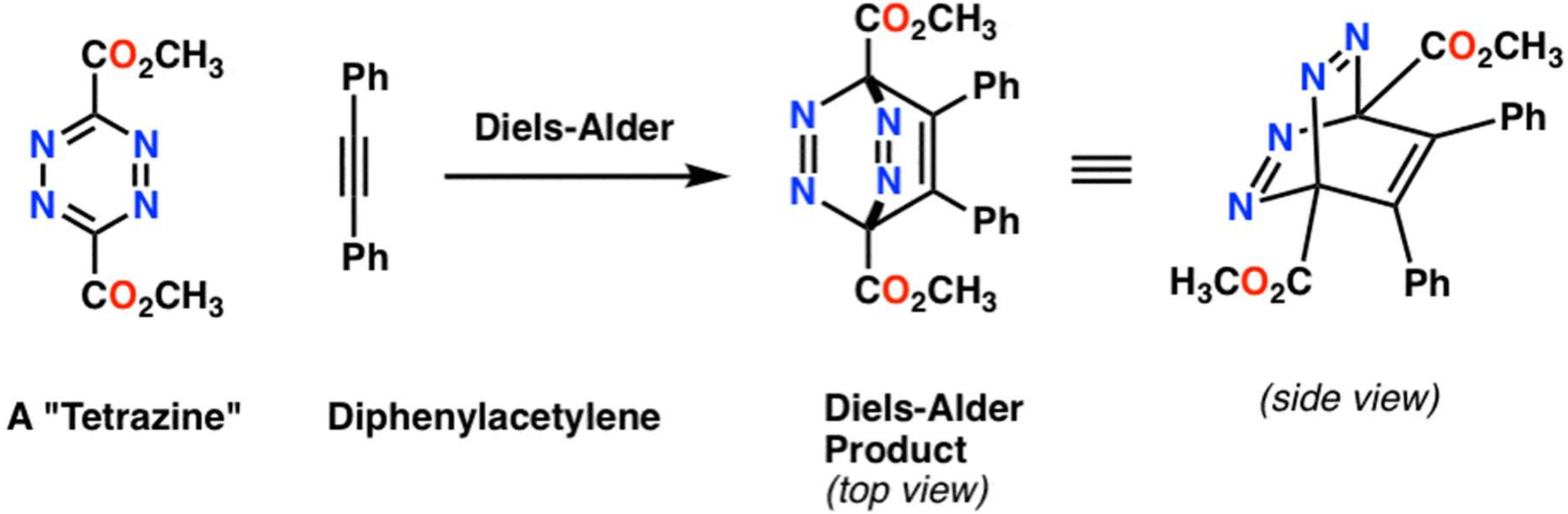
Pyrone and alkyne: in equilibrium
with Diels-Alder product...

...but this *retro*-Diels Alder is **irreversible**

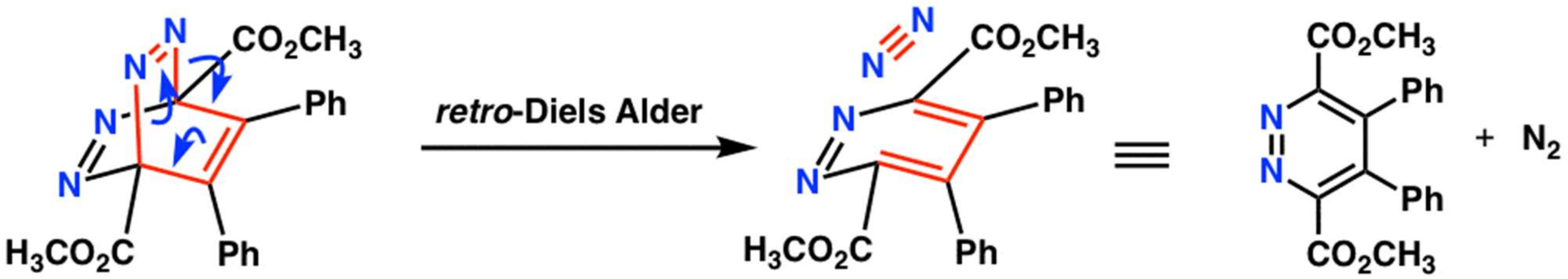



Even if the *retro*-Diels Alder to regenerate the pyrone and alkyne does occur,
it just sets up an equilibrium.

## Another Diels-Alder / Retro-Diels Alder Sequence

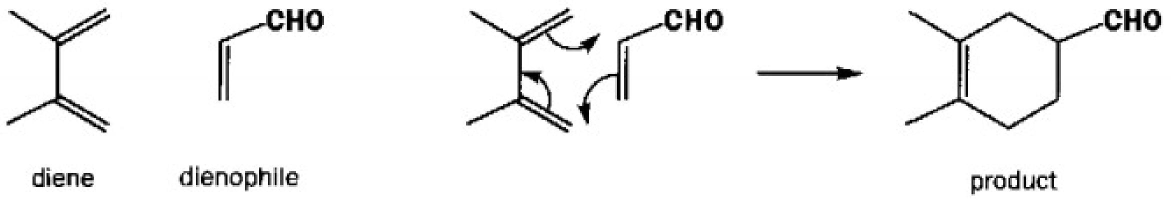


## Heating Leads to A Retro-Diels Alder With Irreversible Loss Of $N_2$

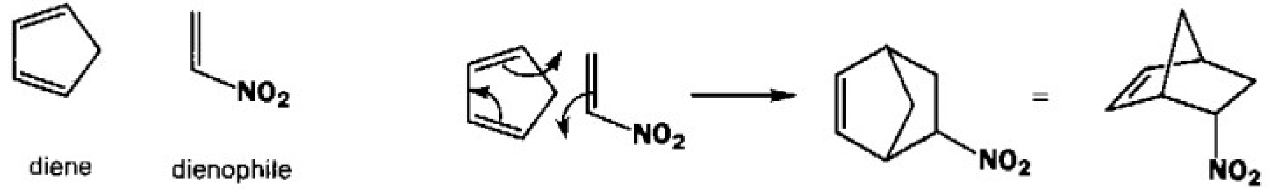


# Example:

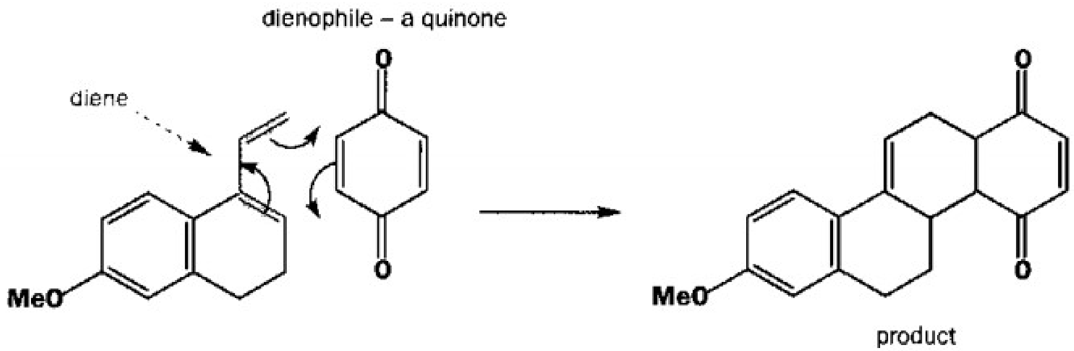
1-


CHO
diene
dienophile
CHO
CHO
product

2-

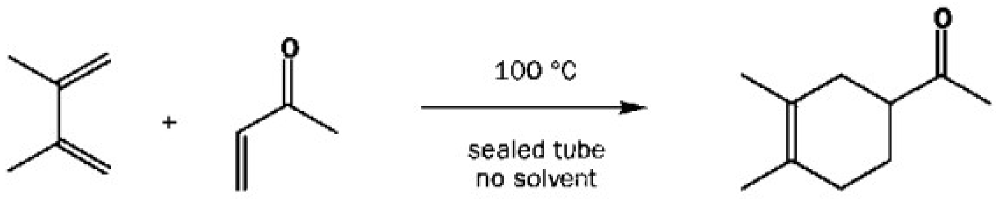

diene
dienophile
NO2
NO2
NO2
NO2

3-

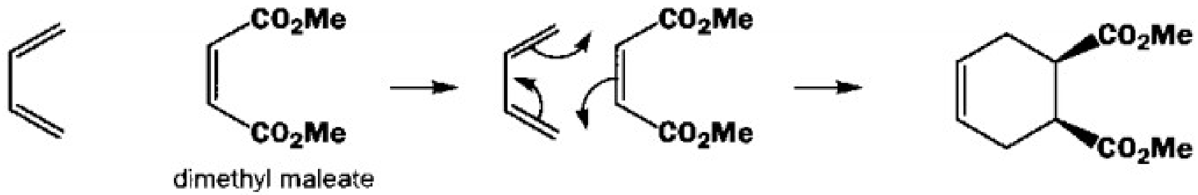

dienophile – a quinone
diene
MeO
O
O
MeO
O
O
product

# Example:

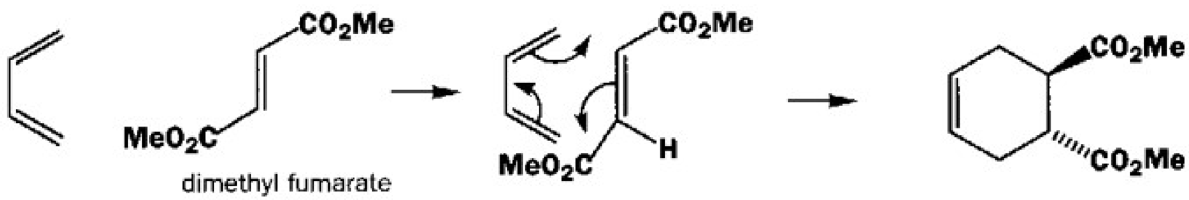
4-


O
100 °C
sealed tube
no solvent
O

5-

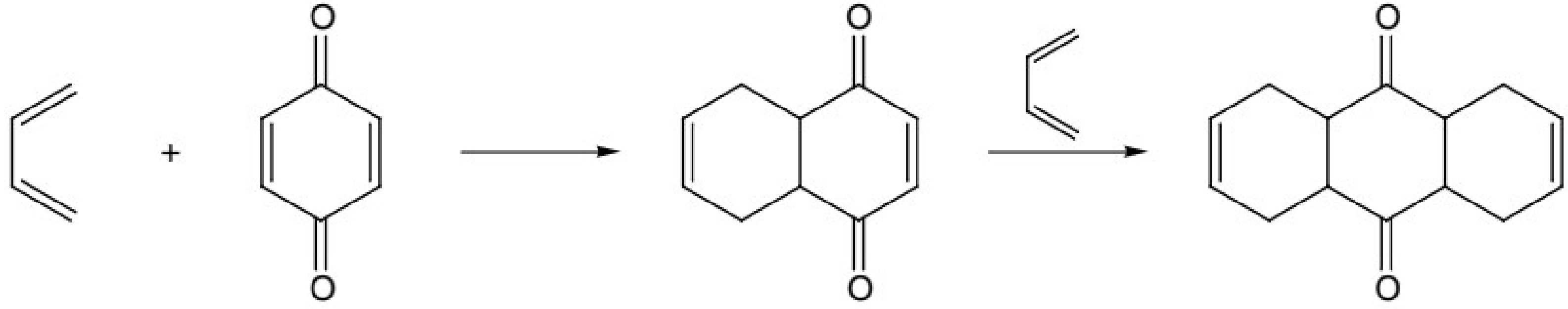

CO2Me
CO2Me
dimethyl maleate
CO2Me
CO2Me
CO2Me
CO2Me

6-

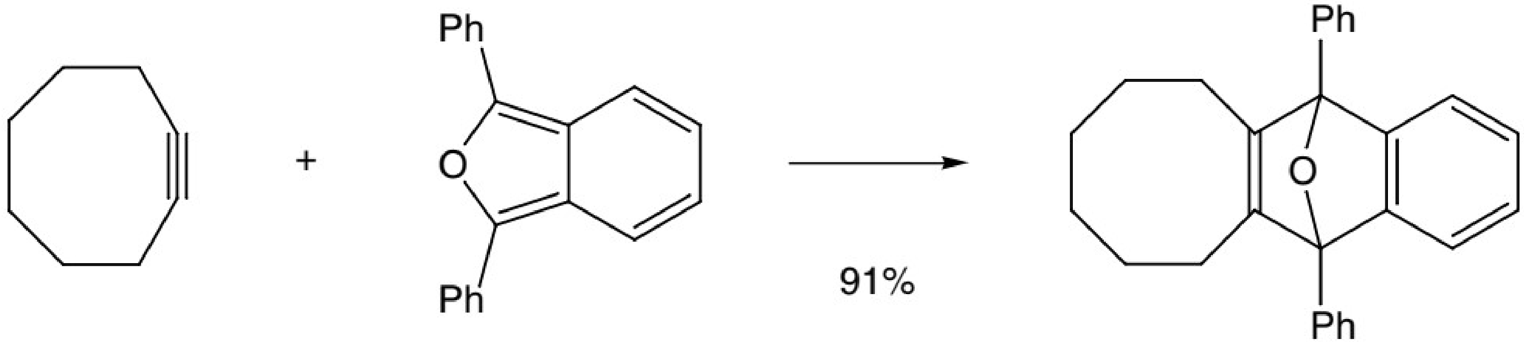

CO2Me
MeO2C
dimethyl fumarate
CO2Me
MeO2C
H
CO2Me
CO2Me

# Example:

7-

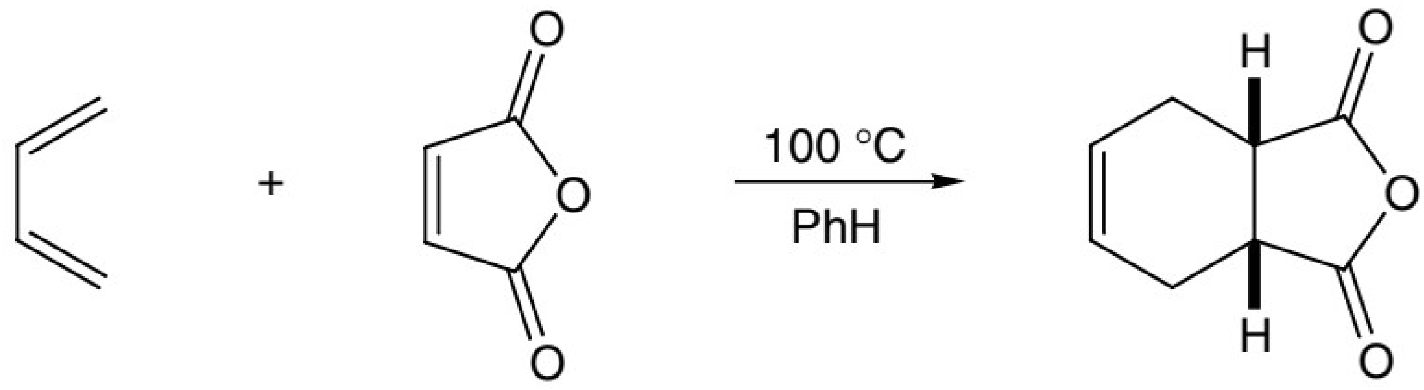


8-

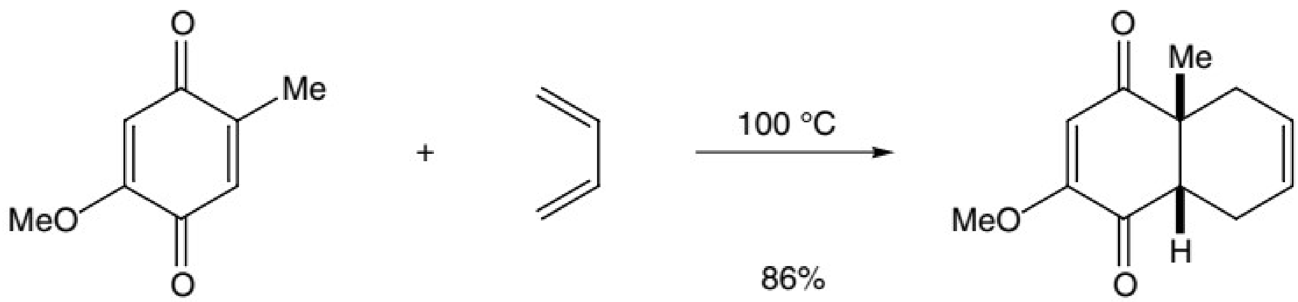


# Example:

9-

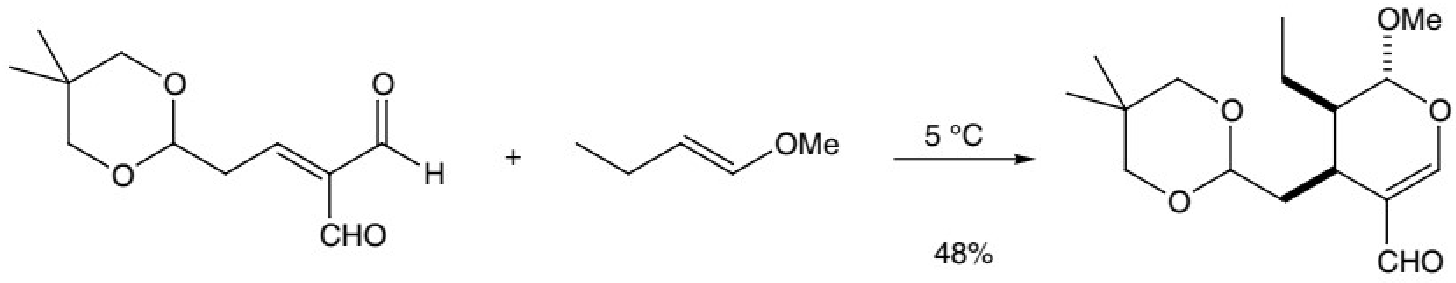



10-

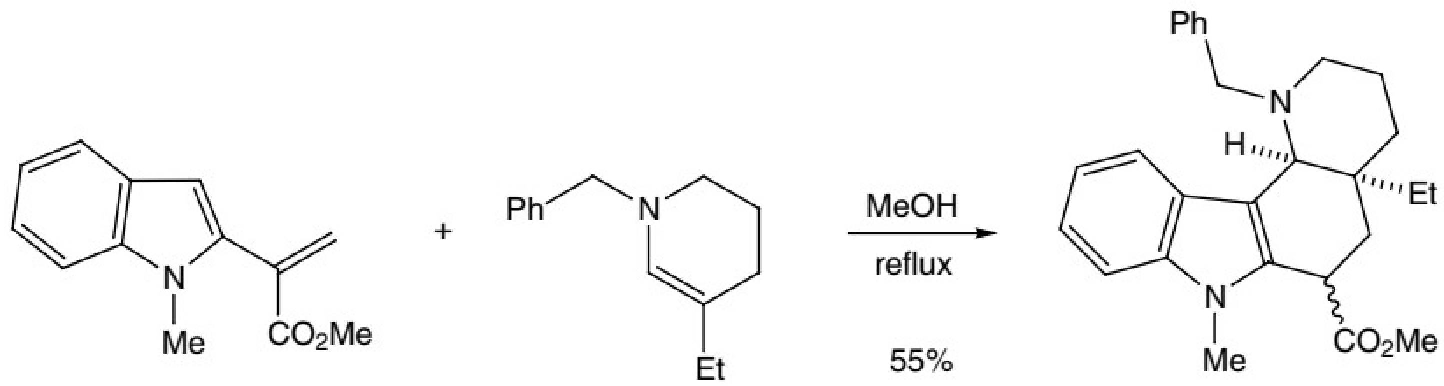


# Example:

11-

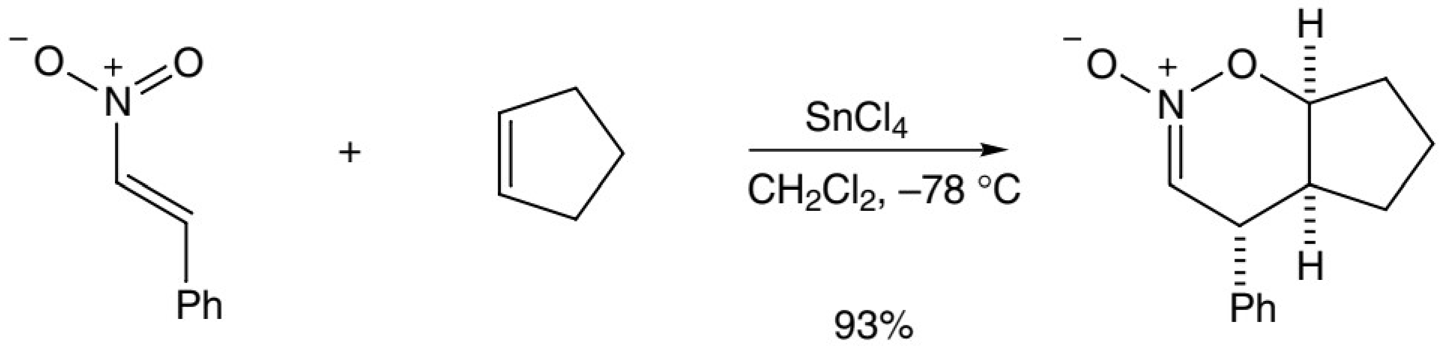



12-

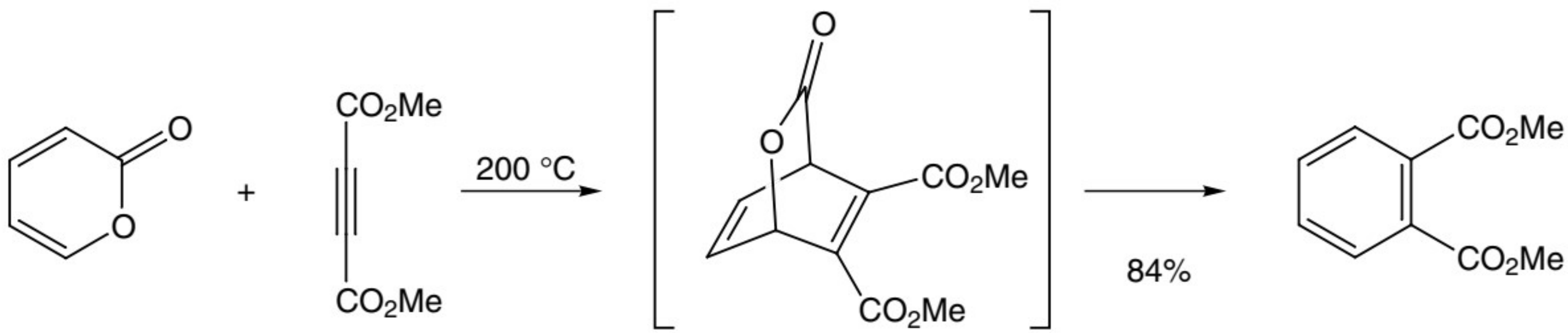


# Example:

13-

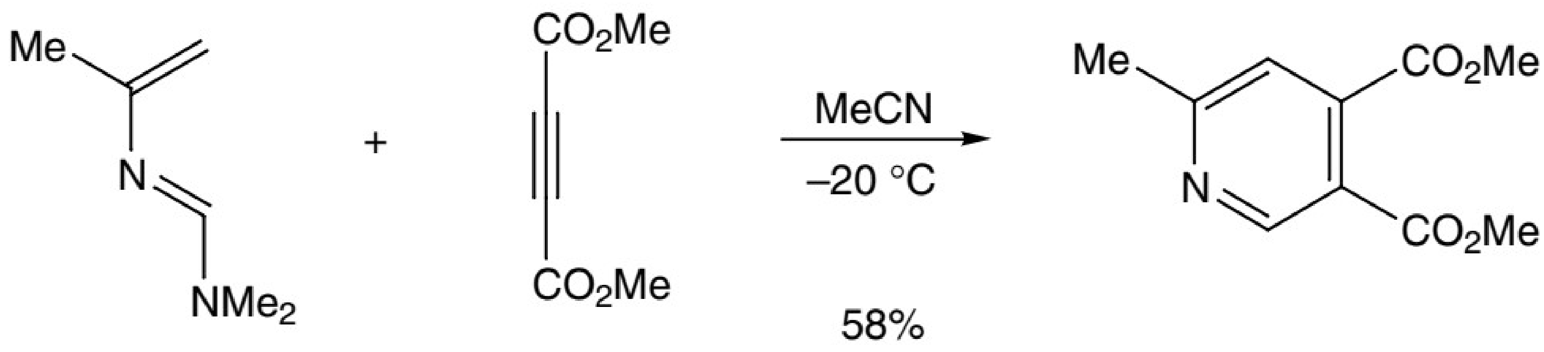



14-

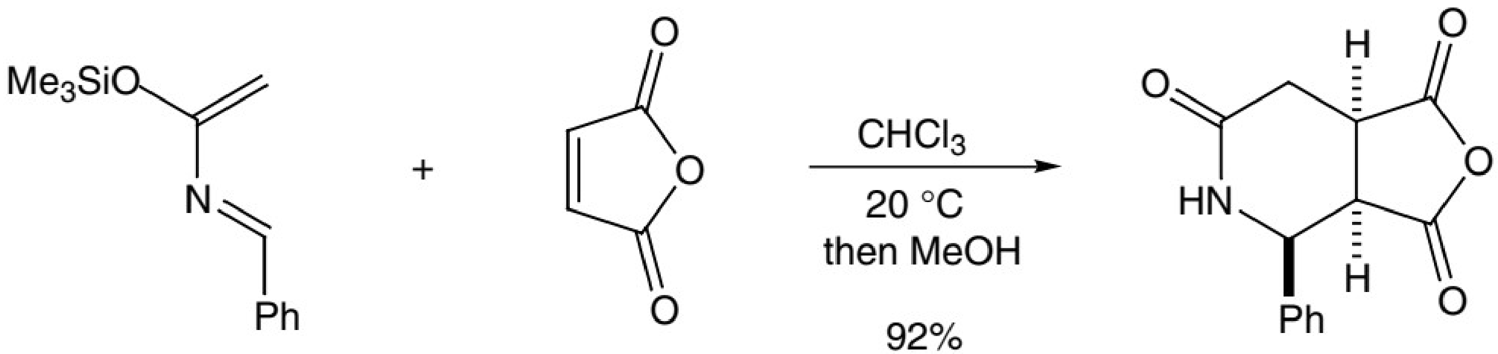


# Example:

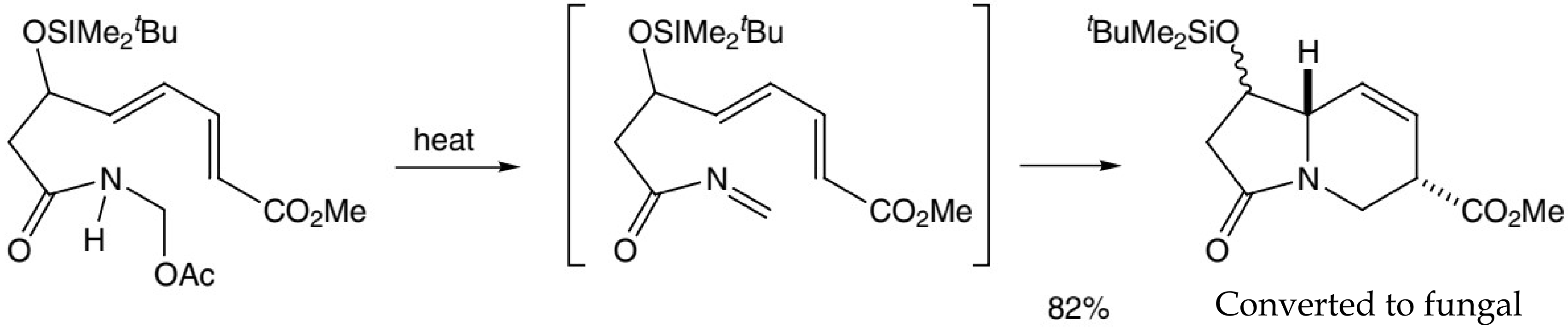
15-



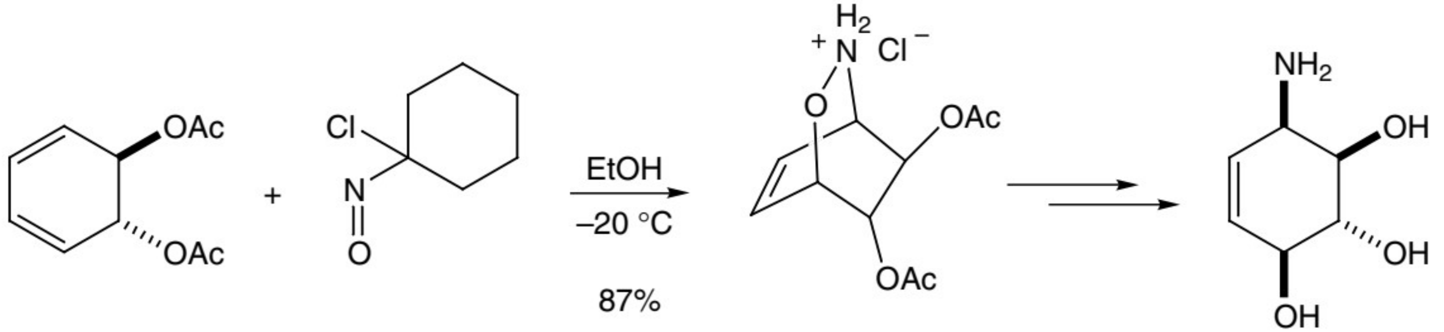

16-



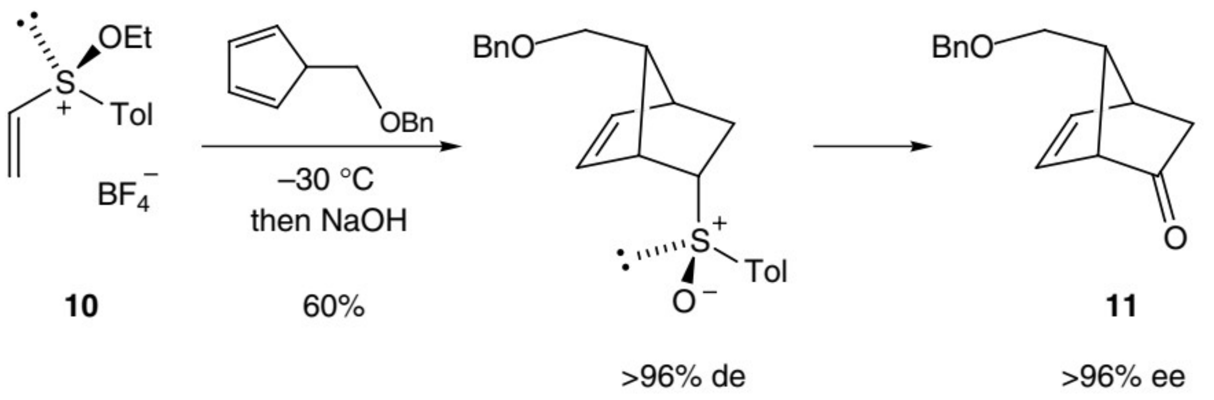
# Application of Diels Alder reaction in synthesis of drugs



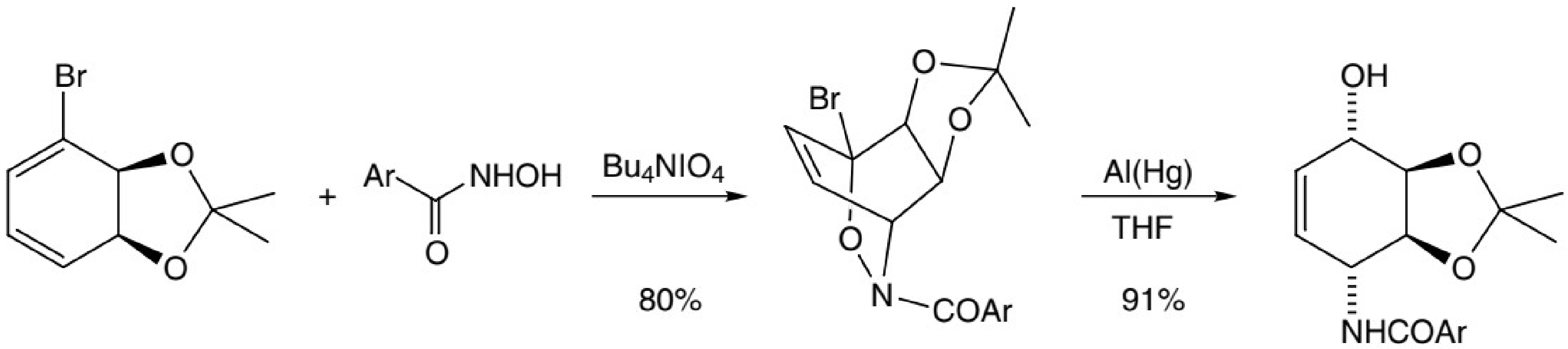

Converted to fungal neurotoxin slaframine



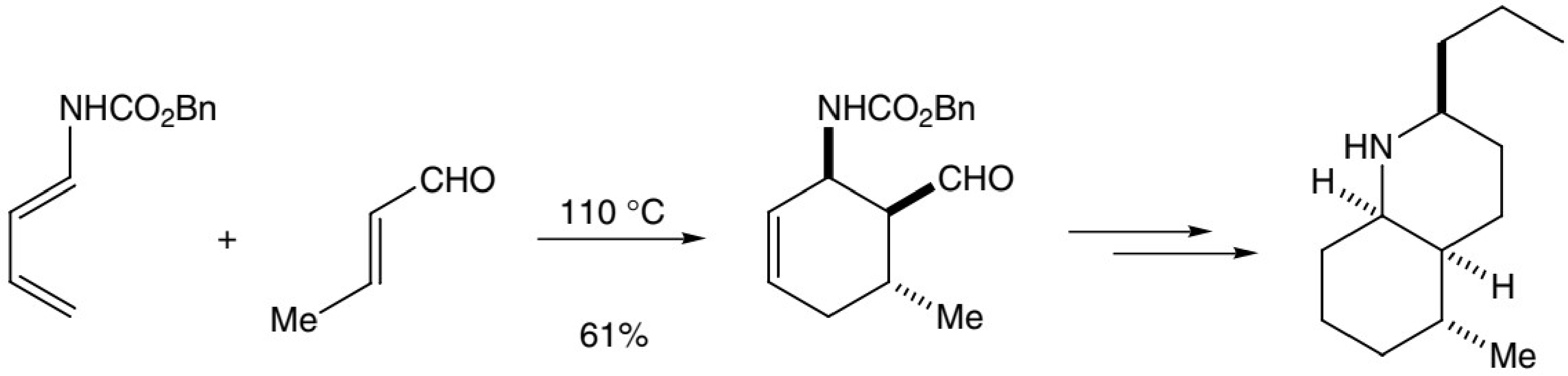

conduramine F1




Used in the synthesis of prostaglandins




Used in the synthesis of lycoricidine



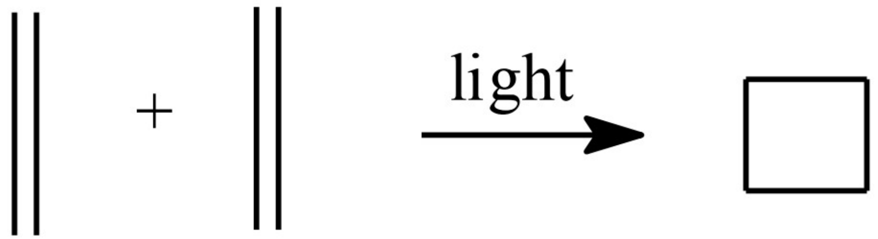

Pumilitotoxine (Poison)

# 1. Cycloaddition Reaction

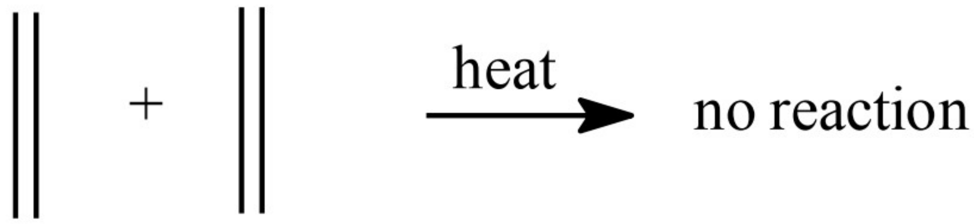
- Formation of a new ring.
- More than one bond form simultaneously.
- No intermediate formation.
- Three important classification of cycloaddition reaction
    I. Diels-Alder reaction
    II. [1,3]-Dipolar cycloaddition
    **III. [2+2] cycloaddition**

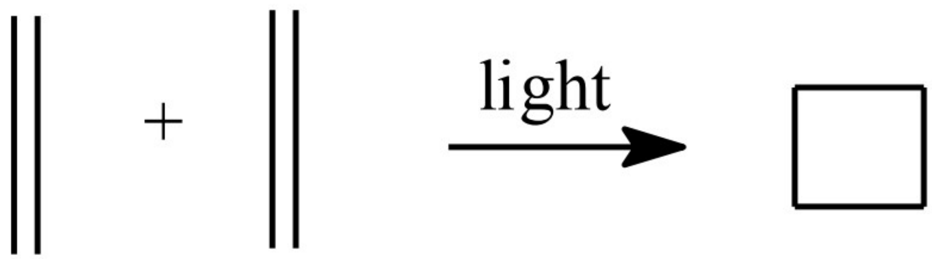
# [2+2] Cycloaddition

- These reaction takes place under photochemical conditions.
- Formation of compounds containing a four membered ring.

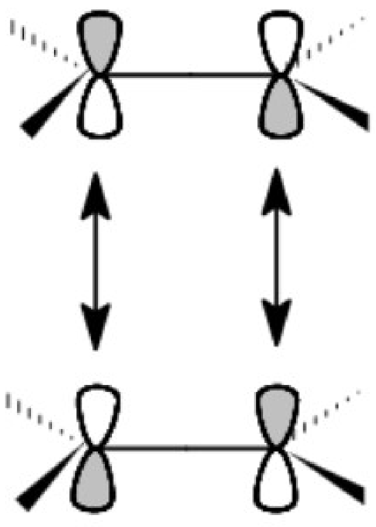


**WHY?**



## Reason:

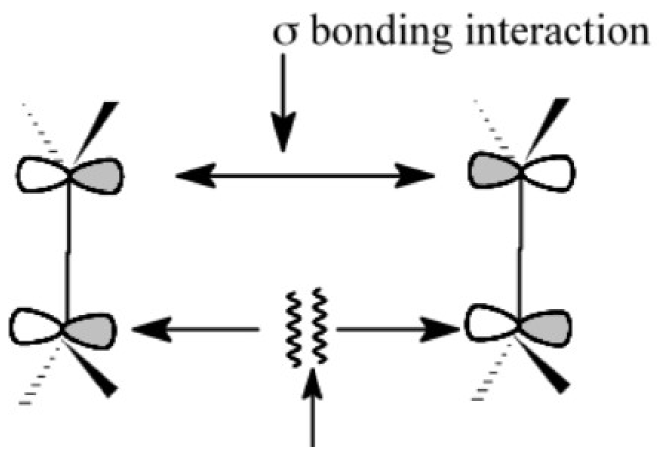


HOMO of excited
state molecule

Two new σ bonds can
Form simultaneously

LUMO of ground
State molecule

LIGHT- reaction




HOMO of
one alkene

σ antibonding
interaction

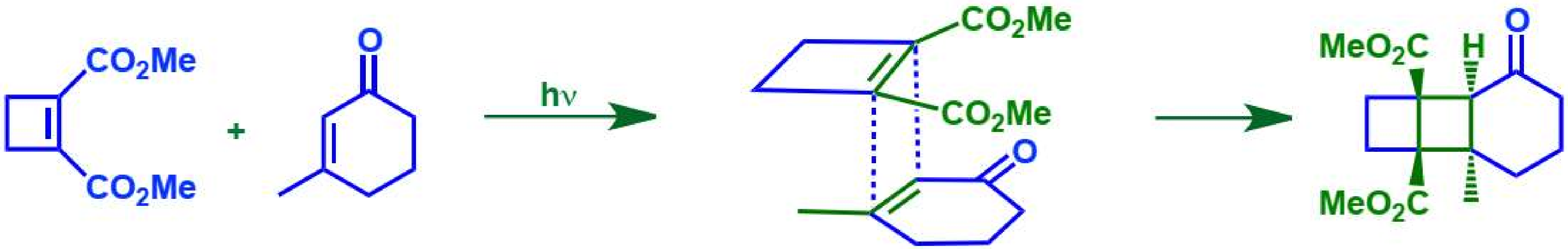
LUMO of one
alkene

HEAT- no reaction

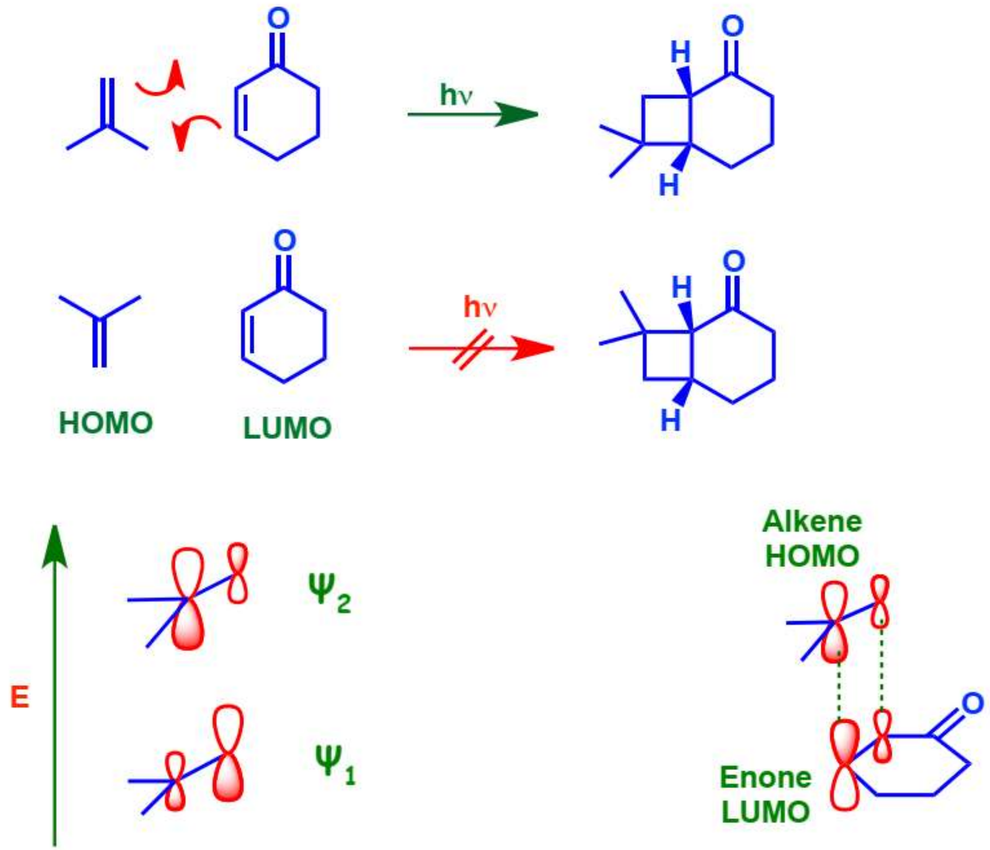
Irradiation *EXCITES* an electron
From the HOMO to the LUMO
So, the excited state HOMO is now
the same symmetry as the ground state LUMO

# [2+2] Cycloaddition is stereospecific.

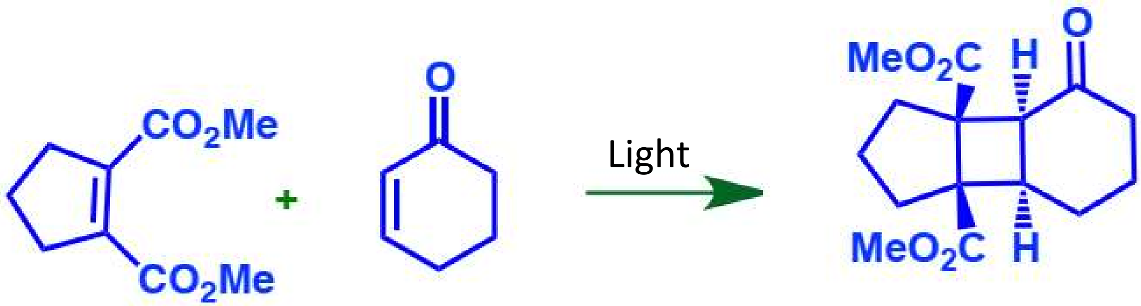
- Less hindered transition is observed



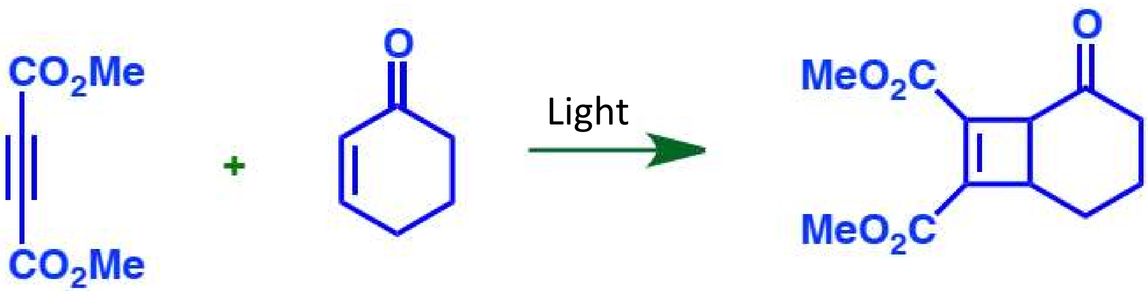
# [2+2] Cycloaddition is regioselective



## Example:

1- 

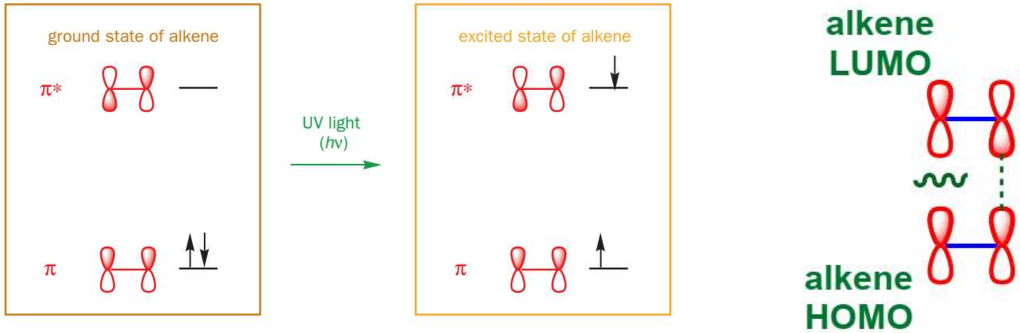
$CO_2Me$
$CO_2Me$
+
O
Light
$MeO_2C$
H
O
$MeO_2C$
H


2- 

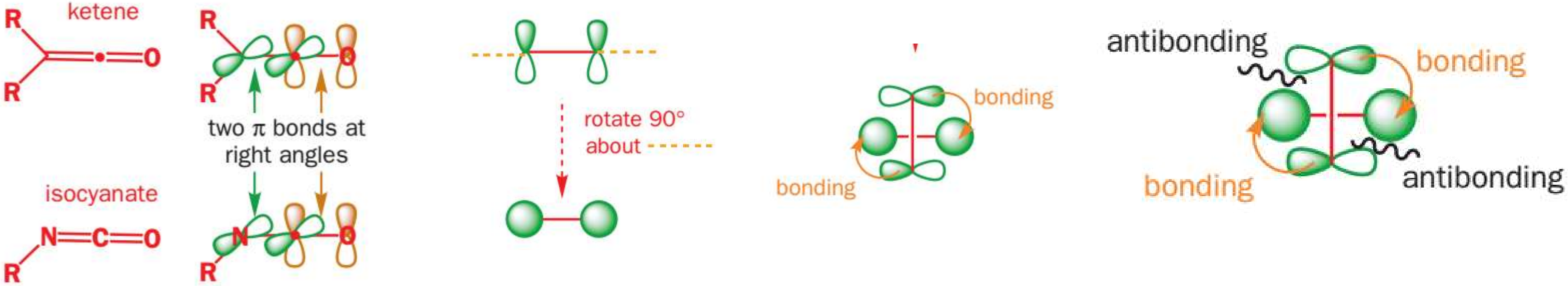
$CO_2Me$
$CO_2Me$
+
O
Light
O
$MeO_2C$
$MeO_2C$

# Thermal [2+2] cycloaddition

- HOMO/LUMO combination is antibonding at one end

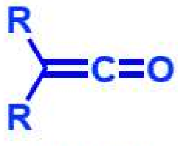



- If HOMO & LUMO approaches perpendicular to each other

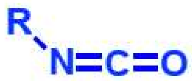


# Condition for Thermal [2+2] cycloaddition

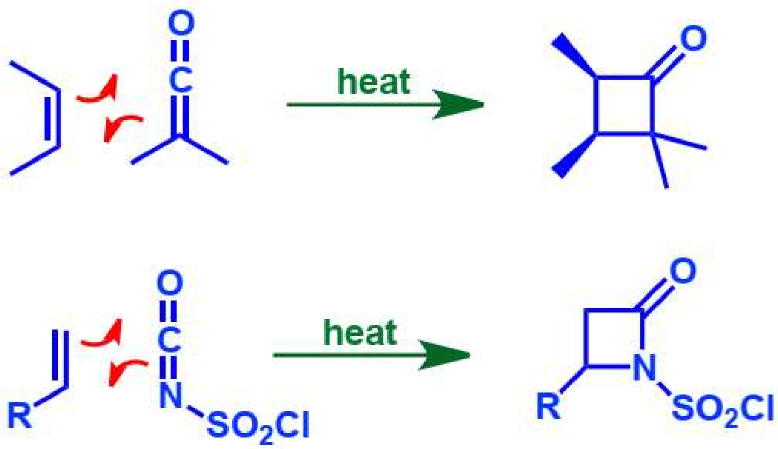
- Two double bond at single carbon.

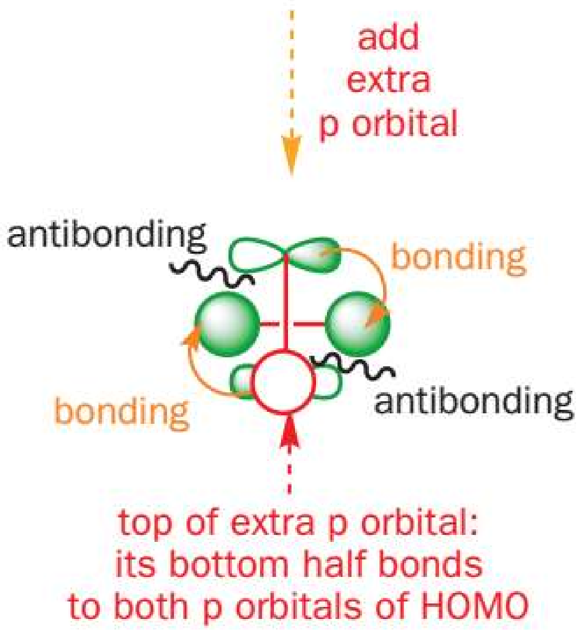


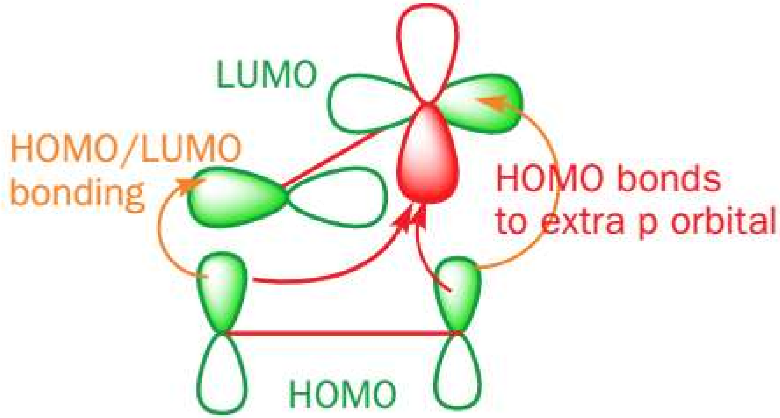
ketene



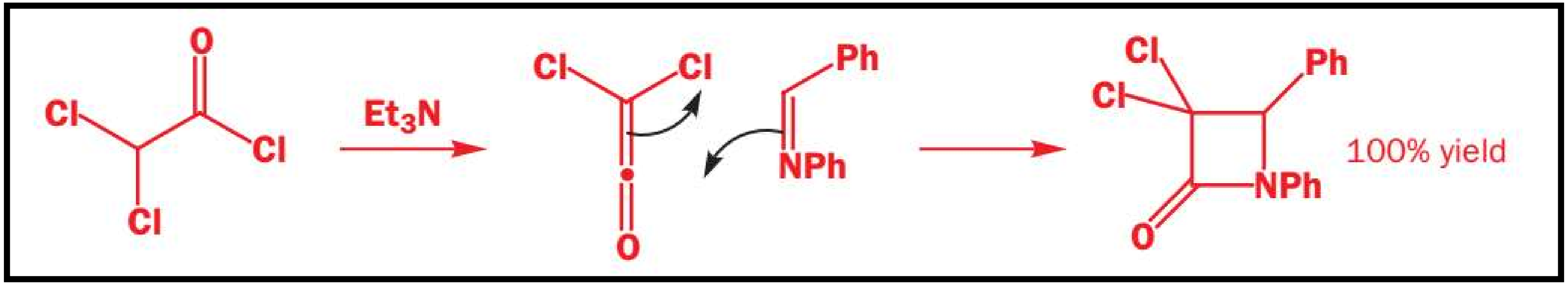
isocyanate

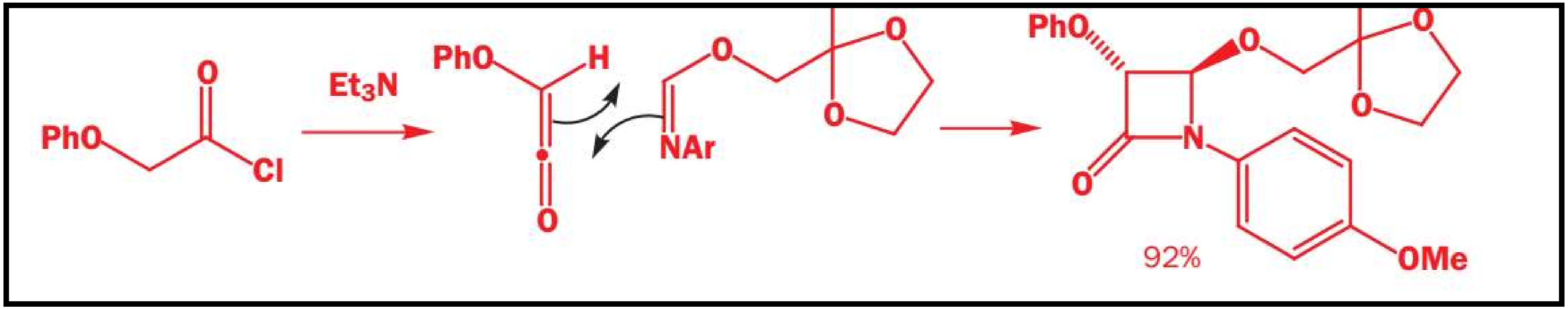


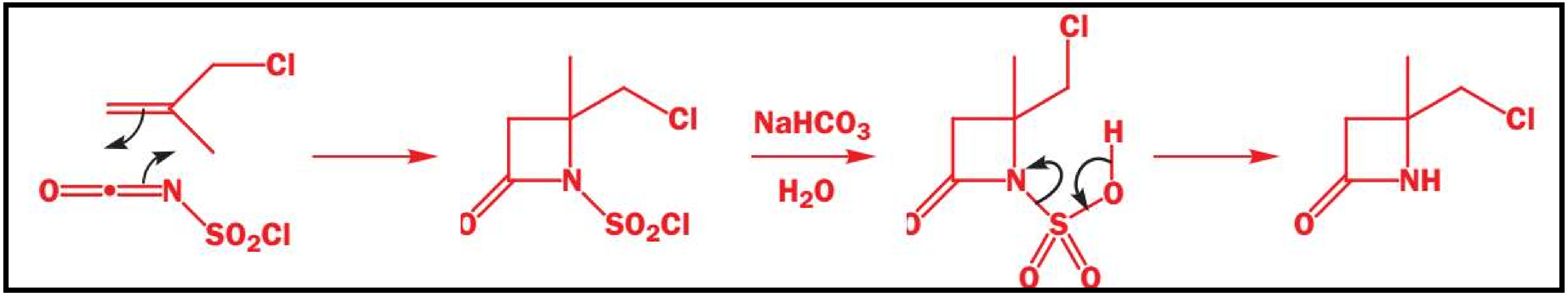



Ketenes have a central sp carbon atom with an extra π bond (the C=O) at right angles to the first alkene—perfect for thermal [2 + 2] cycloadditions. They are also electrophilic and so have suitable low-energy LUMOs.


Et3N
100% yield


Et3N
92%


NaHCO3
H2O